

UFMFUU-15-3:

Final Aerospace Group Design Report

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1. Abstract

The aim of this report is to present an application of a preliminary design procedure for a short-range strut braced airliner for operation on intra-European routes, primarily optimised for minimum environmental impact. A high density 190 Pax jet aircraft cruising at M = 0.82, a 9.5 km altitude, 1000nm design range and entry to service in 2030 is considered. To ensure a thorough and comprehensive development of the aircraft, the report is structured around 10 distinct technical roles. They are as follows:

- Aircraft mass and balance
- Aerodynamics
- Structures
- Power plant
- Aircraft performance calculations over the mission profile
- Aircraft Stability and Flight Control
- Aircraft Economics
- Landing gear design
- Aircraft Systems and integration
- Beyond final design

Each technical role delves into the intricate aspects of aircraft design within its specific area of expertise. The findings of this report affirm the practicality of designing an aircraft characterised by a high aspect ratio and an elevated lift-to-drag ratio through strut braced design. These design features hold promise for significantly reducing fuel consumption, thereby advancing environmental sustainability in aviation.

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1.1 Nomenclature

Abbreviation	Definition
AEA	Association of European Airlines
AFC	Airframe Cost
AOA	Angle Of Attack
BPR	Bypass Ratio
CAGR	Compound Annual Growth Rate
CG	Centre of Gravity
DOC	Direct Operating Cost
FDR	Final Design Report
ICA	Initial Cruise Altitude
ICAO	International Civil Aviation Organization
ISA	International Standard Atmosphere
KIAS	Knots Indicated Airspeed
M	Mach Number
MAC	Mean Aerodynamic Cord
MRW	Maximum Ramp Weight
MTOW	Maximum Take-off Weight
MWE	Manufacturers Weight Empty
MSP	Manufacturers Study price
Nm	Nautical Miles
PDR	Preliminary Design Report
ROC	Rate of Climb

SAF	Sustainable Aviation Fuel
SFC	Specific Fuel Consumption
TAT	Turnaround Time
W	Weight

1.2 List of Symbols

Symbol	**Definition**
w_f	Fuselage Width
k_f	Empirical value for position of ¼ cord of % fuselage length
S	Wing Area
c	Mean aerodynamic cord
a_w	Lift curve slope
T	Thrust
t_p	Distance of thrust line from cg
η	Tail efficiency C
AR	Aspect ratio
l_f	Fuselage Length
w_f	Fuselage Width
l_h	Horizontal tail arm
l_v	Vertical tail arm
V_v	Vertical volume tail ratio
b	Wingspan
dε / dα	Downwash
V_h	Horizontal volume tail ratio
C_L	Coefficient of Lift
C_D	Coefficient of Drag
C_{D0}	Coefficient of Profile Drag
C_{Di}	Coefficient of Induced Drag
e	Oswald's Efficiency Factor
C_m	Moment Coefficient
C_{m_a}	Moment Coefficient Slope

C_η	Yaw Moment
C_{η_β}	Yaw Moment due to Sideslip Angle
S_v	Area of Vertical Tail
S_h	Area of Horizontal Tail
b_{vtail}	Span of Vertical Tail
b_h	Span of Horizontal Tail
C_{th}	Chord Tip of Horizontal Tail
C_{rh}	Chord Root of Horizontal Tail
C_{tv}	Chord Tip of Vertical Tail
C_{rv}	Chord Root of Vertical Tail
a_t	Angle of Attack of Vertical Tail
C_{m_e}	Coefficient of Hinge Moment due to Elevator
H_e	Hinge Moment of the Elevator
H_r	Hinge Moment of the Rudder
C_{m_r}	Coefficient of Hinge Moment due to Rudder
b_1	Curve Slop of Hinge Moment due to Angle of Attack
b_2	Curve Slop of Hinge Moment due to Deflection
q	Dynamic Pressure
G	Gearing Ratio
β	Sideslip Angle
σ	Sidewash Angle
δ_r	Deflection of rudder
a_f	Angle of Attack of Vertical Tail
C_{mac_e}	Mean Aerodynamic Chord of Elevator
C_{mac_r}	Mean Aerodynamic Chord of Rudder
P_e	Stick Force of Elevator
P_r	Pedal Force of Rudder

2. Introduction

In the realm of modern aviation, the drive towards sustainability has become the cornerstone for innovation, particularly in the design of commercial aircraft. The purpose of this Airbus, student lead, report is to design either a turbo fan or turboprop powered large airline capable of operating many intra-European routes, which is primarily optimised for minimum fuel burn and other environmental impact. This report presents a unique opportunity to immerse in the intricate world of aircraft design, laying a solid foundation for a future in aerospace. Furthermore, working collaboratively as a team enhances communication skills, this teamwork experience is crucial in navigating the complexities of aerospace projects, where interdisciplinary collaboration and clear communication are key to successful outcomes.

The key design drivers for this project are multifaceted, low direct operating cost, Airport operations and high operational reliability. The key requirements are to design an aircraft capable of travelling 1000 nautical miles, carrying 190 passengers. The aircraft is set for entry into service by 2030.

To achieve these goals, the report will involve an extensive dive into aircraft preliminary design, broken down into 10 technical roles, using data validation and collection where appropriate. The immediate responsibility of the team involves progressing the initial concept outlined in the Preliminary Design Presentation through a comprehensive performance mission profile, as detailed in Technical Role 9. This critical step is essential to confirm that all the specified design requirements were adequately met. Upon successful completion of this process, a validated design is then established.

After finalising a valid design, the focus shifts to refinement. This stage involves iterating the aircraft's design to achieve an optimised configuration. Subsequently, the optimised design can be thoroughly explored across the various technical roles.

This report will additionally delve into considerations that extend beyond the final design phase.

3. Design Requirements

		U2020
Passenger Capacity (typical airline 1-class HD rules) (4.1)	-	190
Passenger Capacity (typical airline 2-class SR rules) (4.1)	-	Result
Design Range with 1-class passenger capacity (4.1)	nm	1000
Study Mission (4.2)	nm	500
Cruise Speed / Mach (4.3)	kts / -	TBD within range of engine data
Time To Climb (1500ft to ICA at ISA) (4.3)	mins.	≤ 25
Initial Cruise Altitude (@ ISA) (4.3)	ft	TBD within range of engine data
Maximum Cruise Altitude (4.3, 4.10)	ft	TBD, appropriate to ICA
Approach speed (MLW, S-L, ISA) (4.3)	kts CAS	≤ 140
Take Off Field Length (MTOW, S-L, ISA+15) (4.3)	m	≤2000
Landing Field Length (MLW, S-L, ISA) (4.3)	m	≤1600
One Engine Inoperative Altitude (4.4)	ft	Result
VMO / MMO (4.5)	kts CAS / Mach	As calculated
Equivalent Cabin Altitude (at Max Cruise Altitude) (4.10)	ft	8000
Turn-Around Time (4.11)	mins.	Result
Airport compatibility limits (4.11)	-	ICAO Code D taxiing and at gate
DOC target (4.12)	$/seat-nm	Result
Engine availability (i.e. First Flight target) (3.4)	Year	2030

Table 3.1 Design Requirements.

4. Iterations

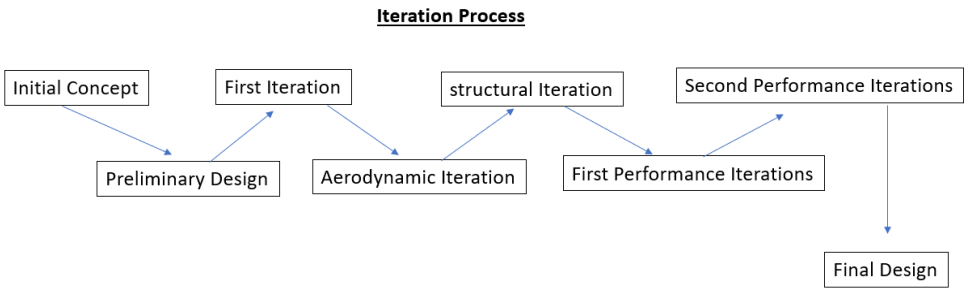


Table 4.1 Iteration Process.

Figure 4.1 shows the iteration process taken before deciding on a final design. After the initial concept and preliminary design, a strut braced high aspect ratio design was chosen (further details section 5), after that the first iteration was investigated along with the aerodynamic iteration and its performance was checked against our design requirements (process detail in section 9) the fuel burnt for our study mission was then found to be 4380kg. A further iteration was found through structural analysis which varied the strut placement to further improve structural integrity of our wings (details section 13). Finally, the key performance parameters were investigated such as cruise

altitude, cruise Mach number and through 9 performance iterations the total fuel burned was refined to 4200kg; this then became the final change which effects the mission profile (results in section 9.1) table 9.1 and 9.2 (details of iteration in section 10.7).

5. Concept Selection

The journey into optimising the final design of the aircraft began with various concepts being produced. Following that, the concepts were down selected with the use of a design matrix to reach to a decision of the final concept. The initial concept designs consisted of a broad range of designs, with the goal to increase efficiency and reduce environmental impacts. Figure 5.1 shows the 6 conceptual designs that were considered for the final approach. The 6 designs are numbered in conjunction with the order of the design matrix.

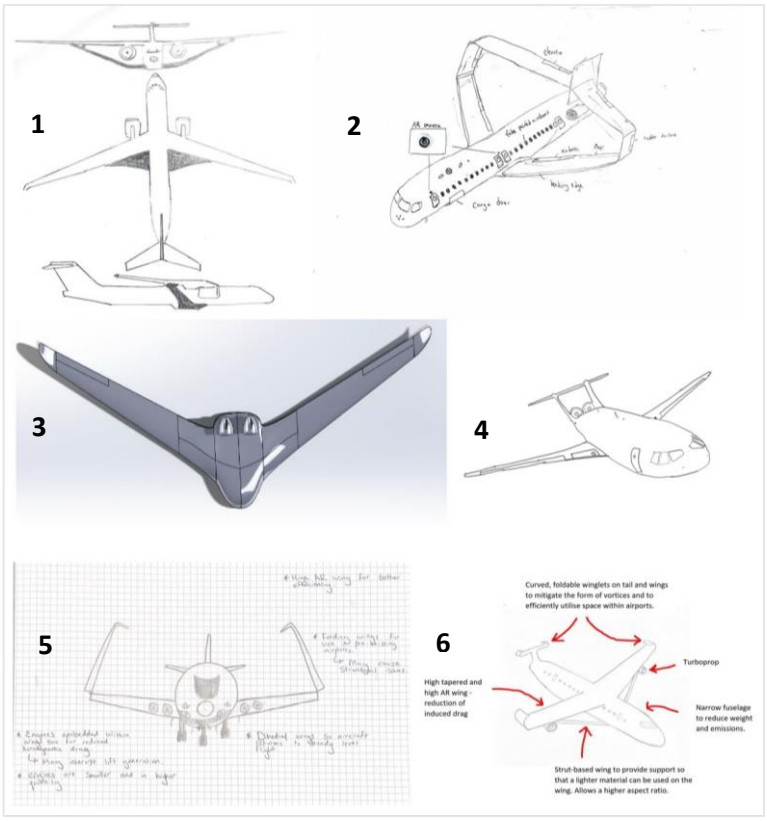


Figure 5.1 Concept Designs Labelled 1 to 6.

Design 1 is a strut-braced turbofan which allows a higher aspect ratio whilst increasing structural efficiency. The wing design was researched to reduce up to 10% of fuel burn.

Design 2 is a box wing windowless fuselage that reduces weight with its statically indeterminant wing structure. The windowless fuselage provides more strength to the fuselage structure, with a potential of less materials being used in production.

Design 3 is a blended wing body that is designed for a significantly high lift to drag ratio. This would reduce fuel consumption and in turn, would lower environmental impact.

Design 4 is a double-bubble plane which provides a streamlined aircraft with its engines being merged into the rear of the aircraft and its low wing position.

Design 5 is an aircraft with folding wing and embedded turbofans. The embedded engines would reduce the induced drag whilst the foldable wingtips would allow it to fit into multiple ICAO codes.

Lastly, design 6 is another strut-based plane with foldable wingtips and turboprop engines. The aircraft has good synergy between the high-wing and struts as lighter weight materials can be utilised to provide a higher aspect ratio wing design.

After an in-depth analysis of the conceptual designs, a design matrix was developed narrow down to a decision. The design matrix would consist of the key design drivers for the final design, in terms of efficiency, environmental impact and costs. Design 1 and 6 showed the most potential for the optimal

concept but there were features on the other designs which showed good synergy between its components. For instance, the large aspect ratio with wingtips would increase overall the overall efficiency. The folding wingtips will be kept from design 5 for the aircraft to function into multiple ICAO codes.

6. Justification of Component Selection

6.1 Fuselage

The fuselage was chosen to be windowless, for its multiple benefits whilst taking into consideration of customer satisfaction. The fuselage will also be narrow bodied with six-abreast seating which will be explained in-depth in the seating arrangement section.

6.2 Wing

The preferred concept for the wing was the use of struts. This would allow high lift generation due to its high aspect ratio. The wings would also have good synergy with the struts, as the high-wing, strut-based design would provide better structural support for the wing, whilst increasing lift capabilities. The high wing is positioned so that there is more room for a high bypass-ratio turbofan as well as keeping it clear from foreign object debris (FOD). The high wing leads to high stability, which is why an anhedral wig was chosen as it would allow for more roll manoeuvrability for the aircraft. The sweep angle is at 25 degrees, as lower angles will provide better performance at lower speeds and short-haul flights. This also would delay the formation of shockwaves.

6.3 Engine

A turbofan engine was chosen over the turboprop as the aircraft will travel at higher speeds (around Mach 0.7-0.8). A turboprop will not be efficient at these speeds as the prop produces shocks. The choice of turbofan would also mean shorter journeys with more flights per day. Over time, this would increase the return on investment of the aircraft. The turboprop may produce high NOx compared to the turboprop but with the reduced time in air, the difference would be minimal. The location of the engines will be below the wing for a better centre of gravity and a reduction of wing bending moments.

6.4 Seating Arrangements

The seating arrangement proposed is a six-abreast seating plan. This would form a single aisle along the fuselage which provides a reduction in weight in comparison to a seven-abreast layout. The seating plan forms a weight-reduced, narrow fuselage leading to fuel efficiency and a reduction of NOx and other harmful gases. An article also mentioned that due to the supply and demand, the narrow-body aircraft market is expected to grow between 2023 and 2030 at a compound annual growth rate of 3%, which is further proof towards the importance of a narrow-body fuselage.

6.5 Landing Gear

We have chosen the retractable tricycle arrangement for our design.

Strut assemblies will be made with Carbon Steel owing to a good balance of strength and weight properties comparing to Stainless Steel.

6.6 Tail

Initially, a cruciform tail was selected for its aerodynamic advantages, but later, a T-tail was preferred to mitigate the engine wake turbulence issues due to the high mounted engines. This design change aimed to enhance overall aircraft stability and flight efficiency. Tail justification is further discussed in the stability and control section of the report.

7. Cabin Arrangement, Cross-Section, and Fuselage Length

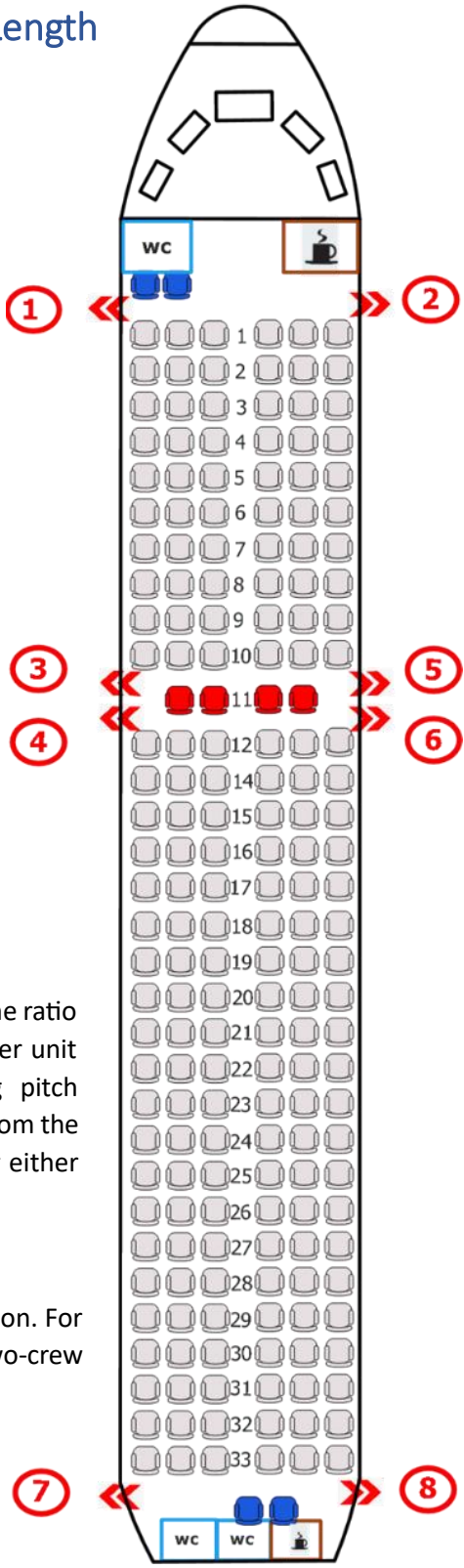
Figure 7.1 and Figure 7.2 are keys for the adjacent Cabin arrangement.

Figure 7.1 Cabin Arrangement Key.



Figure - 7.2 Exit Key





7.1 Fuselage Length

The rear fuselage length is based on data collection from similar aircraft and the ratio of l_{tail}/l_f of 0.25 is chosen, this should also be a house for the Auxiliary power unit (APU). The cabin length was calculated using Airbus economy seating pitch requirements (0.74 meters), the number of rows (32), a reasonable distance from the first row to the cockpit wall and the last row to the rear cabin wall (1 meter either side) and lavatory dimensions suggested by airbus is given by:

$C_{length} = (0.74 * 32) + 0.73 + 1.45 + 2 = 27.86$ meters

As regards for the nose ratio l_{nose}/l_f a value of 0.03 is chosen from data collection. For the cockpit length ($l_{cockpit}$), Raymer suggested a value of 2.5 meters for a two-crew cockpit. Raymer, D.P.R. (1989)

As for the values for the cabin and cockpit, the cabin length forms 72% of l_f. Hence, the fuselage length can be derived as 25.22/0.72 = 38.7 meters.

This is just over a meter longer than the Airbus A320's fuselage length and therefore the method employed above is reasonable.

Figure 7.2 Cabin Arrangement.

7.2 Exits

CS-25 (25.803 & Appendix. J) states that the aircraft must be capable of being evacuated in 90 seconds using only half the side exits on the aircraft. Using this requirement figure 7.1 & 7.2 delineates the type and location of the exits in the cabin. Exits 3-6 require an increased seating pitch to meet the CS-25 requirements, with rows 11 and 12 being unable to recline. Exits 1,2,7 and 8 require 1 attendant per door. Hence the cabin will have 4 cabin crew.

7.3 Lavatories and Galleys

The designated galleys and lavatories for the cabin will be equipped with the Spaceflex V2 by Safran. This system is an integrated galley and lavatory which conserves space, enabling the addition of two extra rows of seating at the back of the cabin. This approach is similar to that adopted by aircraft such as the Airbus A320neo, which also utilises the same system to maximize Pax.

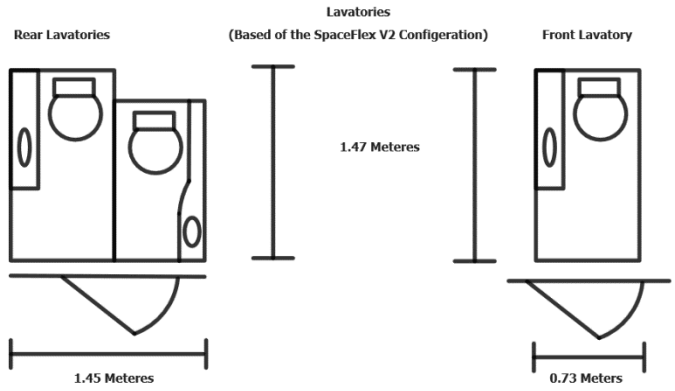


Figure 7.3 Lavatories.

7.4 Cabin Length and the Cross-Section

	1 - Class High Density
Seating	
Seat pitch	**0.71**
Seat Depth	**0.51**
Aisle Width	**0.48**
Minium distance to sidewall for window seat passenger	
Head	**0.03**
Minium standing height	
Aisle	**2.1**
Under side bins	**1.7**

Figure 7.4

Using the number of seats abreast (6), seat width, aisle width, and minimum distance from shoulder to the window, from the airbus standards shown in the table adjacent, the internal trim to trim of the aircraft cabin is calculated:

$D_{INTERNAL} = (2 * 1.52) + (0.03 * 2) + (0.48) = 3.58$ m

The fuselage diameter was determined using structural thickness standards prescribed by Raymer and is given by:

$C_{THICKNESS} = 0.02 * D_{INTERNAL} + 1 = 0.02 * 140.94 + 1 = 0.097$m

Therefore, the external diameter of the fuselage is obtained as:

3.58 + (0.097*2) = 3.77 meters

From data collection, the underfloor cargo container chosen.

is the LD3-45W.

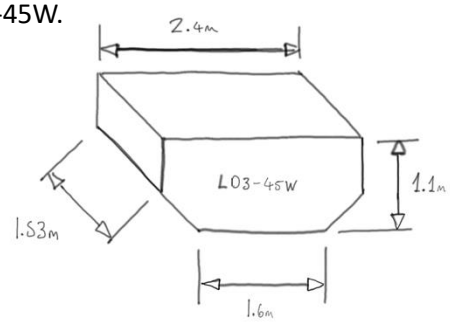


Figure 7.5

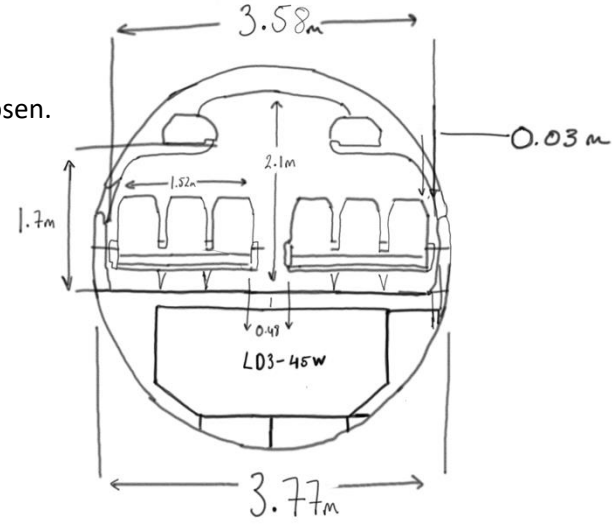


Figure 7.6

8. Mass & Balance

8.1 Initial Weight Estimations

The first method employed to determine the MTOW was a comparison of historical data for aircraft MTOW vs Pax. A value of 82,000 Kg was found for an aircraft with 190 Pax. This is outlined in the figure below.

Next an iterative guess method was employed, and a value of 74,000 Kg was found.

The last method used was a basic weight build up, where fuel weight was estimated from our 1000 nm design range. This value of MTOW was 64,000 Kg

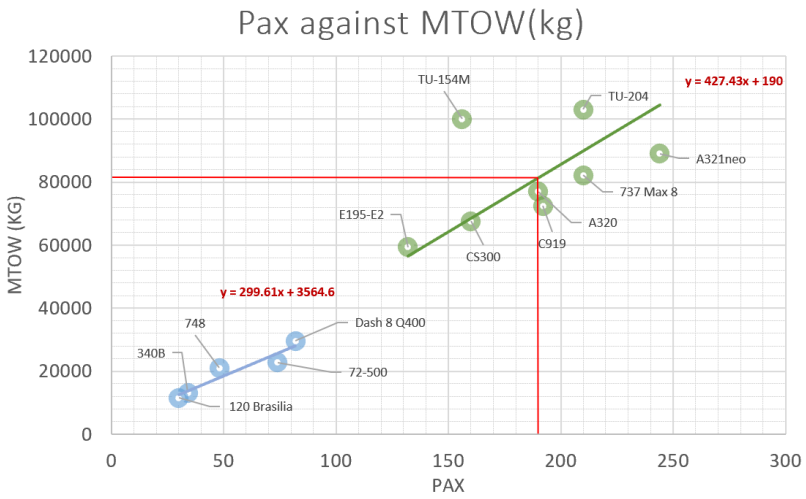


Figure 8.1 Pax against MTOW.

8.2 Revised Weight Build-Up for the Strut based Aircraft and Methodology Validation

Amongst all the design modules employed in aircraft design process, weight module is the most significant one. Evaluating aircraft performance is dependent on a suitable aircraft weight to carry out its intended mission. In interactive design process, the weight design engineers usually follow one published methodology such as that proposed by Roskam or Torenbeek. Al-Shamma, O. and Rashid, A. (n.d.).

There is a drawback of such methodologies however, due to their limited accuracy when being applied to the vast variation of civilian aircraft.

Hence, a new method was employed to the interactive design process. This method suggests the use of many difference equations across different methodologies to each aircraft component.

As a start point the Raymer, reference 3, weight build up method was solely used to calculate an estimate operating empty weight (EOW). This EOW was then compared to the actual EOW of the Airbus A321, and a margin of error was determined, this margin of error was 25%. Other methodologies including Nicolia, Torenbeek, Harris and Kroo were then implemented as an attempt to reduce the margin of error. The margin of error was reduced to 12% and therefore the confidence in the method was enhanced and the method was employed on the conceptual strut-braced aircraft. The A321 was used for validation for its competitive likeness.

The table below shows, for the strut-based aircraft, a breakdown of the component, their weight and the method used to calculate them.

Strut-Based Weight Build Up		
Component	**Calculated Weights Kg**	**Method**
Fuselage	**8036**	**Nicolia**
Wing	**11879**	**Raymer**
Vertical Tail	**1637**	**Torenbeek**
Horizontal Tail	**2456**	**Torenbeek**
Main Landing Gear	**2444**	**Torenbeek**
Nose Landing Gear	**431**	**Torenbeek**
Engines (total)	**6320**	**Harris**
Fuel system	**632**	**Raymer**
Flight Controls	**1611**	**Raymer**
APU	**142**	**Kudu**
Hydraulics	**609**	**Raymer**
Avionics	**827**	**Raymer**
Electronics	**1319**	**Raymer**
Air con, anti ice	**973**	**Torenbeek**
Furnishings	**3769**	**Raymer**
Pain and miscellaneous	**388**	**Torenbeek**
Crew	**418**	**Torenbeek**
Attendants	**749**	**Torenbeek**
Oxygen system	**258**	**Torenbeek**
Operating Items	**1637**	**Torenbeek**
Weights		
EOW	**46535**	
Fuel	**10000**	
Payload	**19950**	
MTOW	**76485**	
MZFW	**71384**	

Table 8.1 Strut Braced Weight Build up.

Remarks:

Note that in this stage the fuel used was estimated based on historical data on fuel weight in comparison to range and a value of 10000kg is chosen.

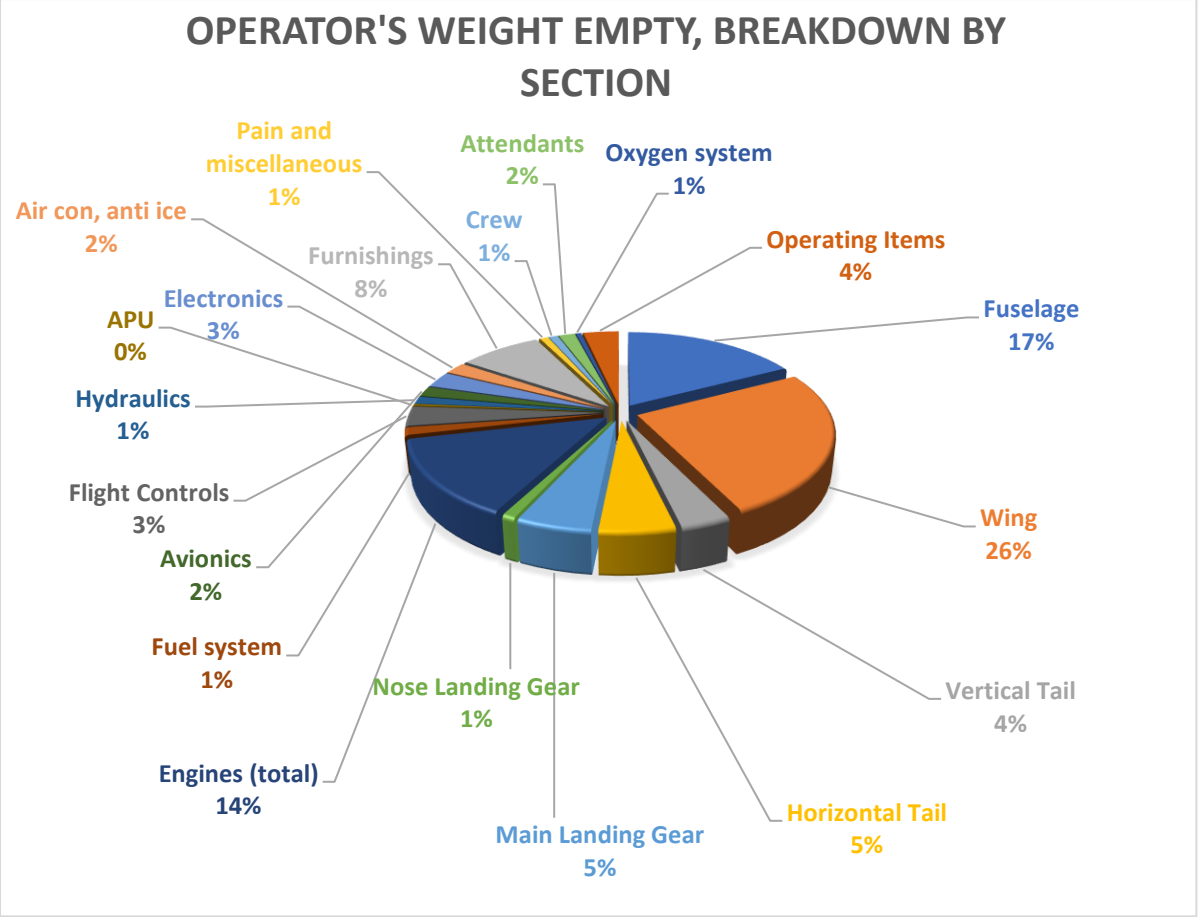


Figure 8.2

8.3 Tail arms

Tail arm is the distance between the aerodynamic centre of the wing and the aerodynamic centre of the horizontal and vertical tail. The values of the tail arm are chosen based data collection of similar aircraft as:

lh = 36 % of lf and lv= 33% of lf

lh = 0.36 * 38.7 = 13.9 meters

lv = 0.33 * 38.7 = 12.8 meters

8.4 Weight component arms, CG positions and Xle

The following CG location assumptions are taken from Tulapurkara's revised example of airplane preliminary design procedure for a transport jet. The CG locations of wing, horizontal and vertical tail, it is assumed that the CG is at 40% of the respective MAC. The fuselage CG is taken to be at 42% of its length. The engine CG is taken to be 40% if its length. For this purpose, the distance of the engine CG from the root chord is measured for various aircraft and 2 m is chosen. All other components (equipment, furnishings etc.) are assumed to have their combined CG location at 42% of the fuselage length. Nose and main landing gear CG positions are taken at 17% and 55%, respectively. The various components are tabulated below.

Using the equivalent trapezoidal wing shown in the figure below. The aerodynamic centre and CG of the wing is 5.55 m and 5.25 m, respectively, behind the leading edge of the root chord of the wing.

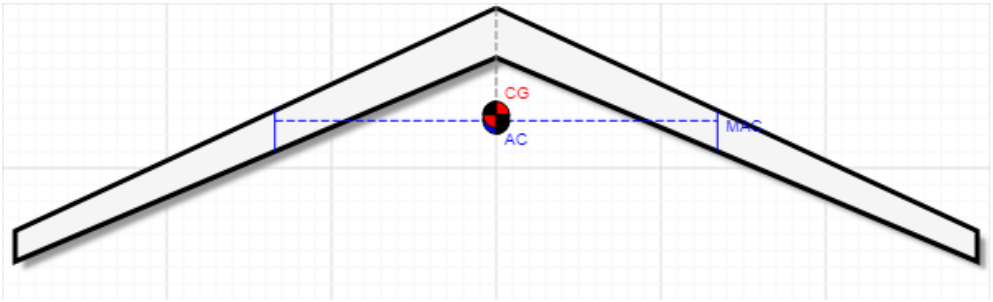


Figure 8.3

The longitudinal location of wing is decided based on the consideration that the CG of the entire aircraft with full payload and fuel is around the quarter chord of the MAC of wing.

Then applying moment equilibrium about the nose of the airplane, the distance of the leading edge of root chord of the wing from the nose (Xle) is calculated to satisfy the aforesaid requirement. The steps to obtain Xle are given below.

Using the a.c data above and Tulapurkara's method for calculating the CG locations (15% behind it's a.c) of the vertical and horizontal tails behind the leading edge of the root chord of wing, the CG values for the vertical and horizontal tail are 18.25 m and 19.45, respectively.

Note that for the Fixed Equipment below the auxiliary power unit, flight control systems, instruments, and navigation equipment. hydraulic systems, electrical systems, avionics systems and furnishing are added together and taken at 42% of the Fuselage length.

Component	Weight (kg)	CG location from nose (m)
Wing	11876	Xle + 5.25
Fuselage	8036	16.25
Horizontal tail	1637	Xle + 19.45
Vertical tail	2456	Xle + 18.25
Engine group	6320	Xle + 2
Nose wheel	431	6.579
Main landing gear	2444	21.285
Fixed equipment total	12157	16.25
Fuel	10000	Xle + 5.55
Payload	19950	17.42
Gross weight	75318	Xle + 5.55

Figure 8.4 Component Weights and CG.

Applying moment equilibrium about the nose of the aircraft, Xle is obtained as 12.1 meters. Consequently, the location of the CG of the aircraft from the nose is at: 12.1 + 5.55 = 17.6m.

8.5 CG travel in critical cases

The movements of the CG under various loading conditions are examined below. The method follows Tulapurkara's example of an airplane preliminary design procedure, reference 4, page 63-64.

8.6 Full payload and no fuel

An assumption has been made that the fuel tanks are located such that the CG of the fuel is at the quarter chord of MAC of wing. Since the CG of the entire airplane is also at the quarter chord of wing MAC, there is no shift in the CG when the fuel has been consumed. Hence, the C.G shift is 0%.

8.7 No payload and no fuel

Since the CG of payload is not at the CG of the entire aircraft, the CG is bound to shift by a certain amount in this case. The moment calculations are performed, and the new CG location is obtained at 17.8 m. Therefore, the CG shift: is 17.8 – 17.6 = 0.2 m or 6.9 % of the MAC.

8.8 No payload and full fuel

For this case, since there is no payload, the CG shifts. On performing calculations, the new CG location is obtained at 17.7 m. Therefore, the CG shift is: 17.7 – 17.6 = 0.1 m

Hence, the CG shift is + 4.1% of the MAC.

8.10 Summary of CG calculation

Wing location (leading edge of root cord of equivalent trapezoidal wing) : 12.1 m

CG of the aircraft from the nose is at: 12.1 + 5.55 = 17.6m.

CG location with full payload and full fuel: 14.61 m

CG travel for no payload and no fuel: + 6.9 % of MAC

CG travel for no payload and full fuel: + 4.1 % of MAC.

According to Raymer, Reference 3, a total CG shift of 15% is acceptable for commercial aircraft. This would suggest that that the aircraft is statically stable.

9. Engines

9.1 Engine choice

To ascertain the suitable engine type for the conceptual aircraft, the passenger capacity of 190 must be considered. Figure 9.1 illustrates that this capacity aligns with existing turbofan aircraft. This alignment serves as the primary rationale for opting for a turbofan engine for our aircraft. While a turboprop engine is a viable option, employing four turboprop engines may be necessary to achieve adequate lift with a reasonably high Maximum Take-off Weight (MTOW). This would compromise the efficiency of the aircraft, a compromise deemed unwarranted given the cruise speed at which turboprops typically operate.

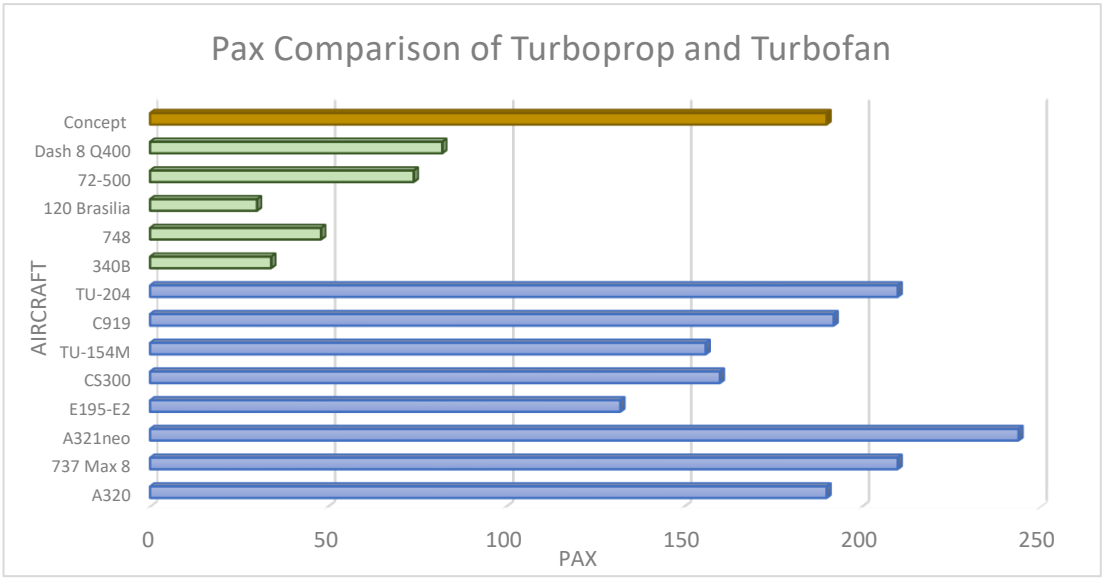


Figure 9.1 Pax Comparison of Turboprop and Turbofan.

Another factor to consider is the cruising speed of the aircraft. A faster aircraft would mean that the block time of an aircraft would be shorter, providing an increase in yearly utilisation. Considering similar Intra-European aircrafts such as the A320 show that these aircrafts travel at a Mach number of around 0.77 and 0.79. To compete with these existing aircrafts, the concept aircraft should travel at similar speeds to gain an advantage in utilisation, whilst limiting fuel consumption.

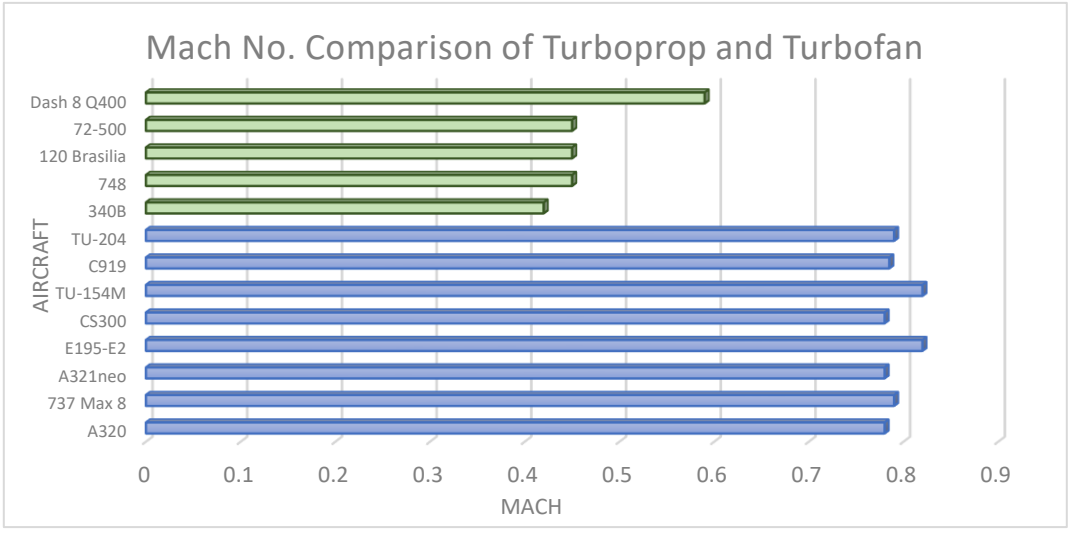


Figure 9.2 Mach Number Comparison of Turboprop and Turbofan.

9.2 Engine Sizing

Engine sizing

For engine sizing, the thrust required per engine was determined at three distinct pinch-points for each iteration. The maximum thrust identified was utilized to scale the baseline turbofan engine.

1) The following equation was used to calculate emergency climb-out with one engine inoperative (RC50) :

$$T = D + Mgsiny$$

The climb gradient at this pinch-point Is 2.4%.

2) This equation was used to calculate top-of-climb, at maximum climb thrust rating (RC40):

$$Fn/Mcl = (L/D)\text{^}(-1) \ + \ (300/\mathrm{V}.101.3)$$

3) Finally, the balanced field take-off distance was plotted against its thrust requirements as shown in figure 9.3:

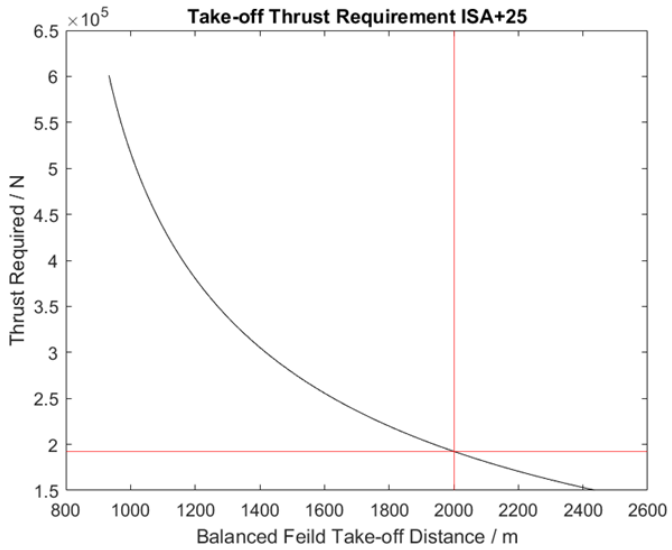


Figure 9.3 thrust requirement vs balanced field take-off distance.

The final engine was sized according to its greatest thrust requirement during the flight profile, take-off, A balanced field take-off at ISA+25 was used.

Figure 9.3 used take-off distance equations from Performance section:

A Thrust requirement of 192,360 N was found to achieve a clearance of 35ft at the end of the 2000m runway. A cautious Static Thrust was then chosen 10% greater than this requirement resulting a sea level static thrust of 211,600N.

The thrust is used to calculate the scale factor and the compnent masses and specifications are acquired in Tables 9.1. and 9.2.

Component Masses

Component	Base mass (kg)	Scaled mass (kg)
Bare Engine mass	2927	2482
Pod systems	1317	1117
Pylon system	732	620
Total	4976	4220

Table 9.1 Final Component Masses.

The total weight of each engine decreased by 756 kg.

Component Specifications

Component	Base Dimensions (m)	Scaled Dimensions (m)
Podded engine overall length	5.5	4.90
Maximum pod diameter	2.45	2.18
Distance of cg aft from bare engine face	1.4	1.24

Table 9.2 Component Specifications.

The overall dimensions of the engines decreased compared to base engine used for scaling.

9.3 Specific Fuel Consumption

Altitudes, thrust requirements, and Specific Fuel Consumption (SFC) were derived by scaling the data obtained from engine specifications. Figure 9.4 illustrates the relationship between altitude and both thrust and SFC at Mach 0.82.

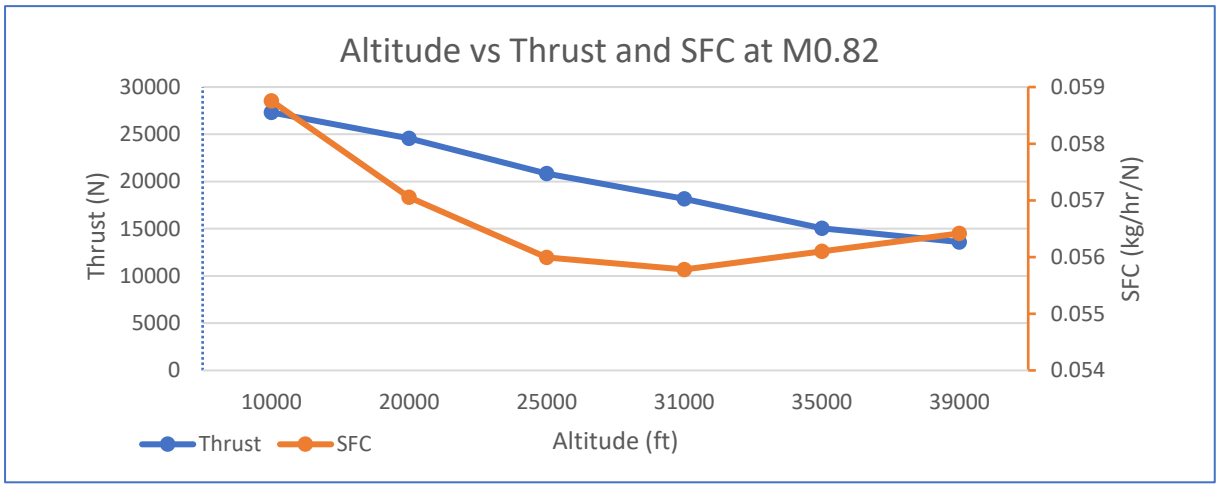


Figure 9.4 Altitude vs Thrust and SFC at M0.82.

The variation in altitude induces changes in pressure and density, leading to adjustments in both thrust and Specific Fuel Consumption (SFC). These alterations are a consequence of the evolving atmospheric conditions as an aircraft ascends or descends to different altitudes. The ideal altitude for cruise at Mach 0.82 is 31000 ft, having the lowest SFC at the same speed.

10. Performance

Equations from performance section are (Anderson, 2021), (take-off, climb, descent, landing approximations)

10.1 FAR Domestic Flight profile

A customised variation on a "FAR Domestic" generic flight profile is used for all performance calculations and the results are compared against the hard performance requirements at every iteration, our study mission is a 500nm mission and this is the more likely mission of our aircraft and a 1000nm mission Is more repetitive of our upper range.

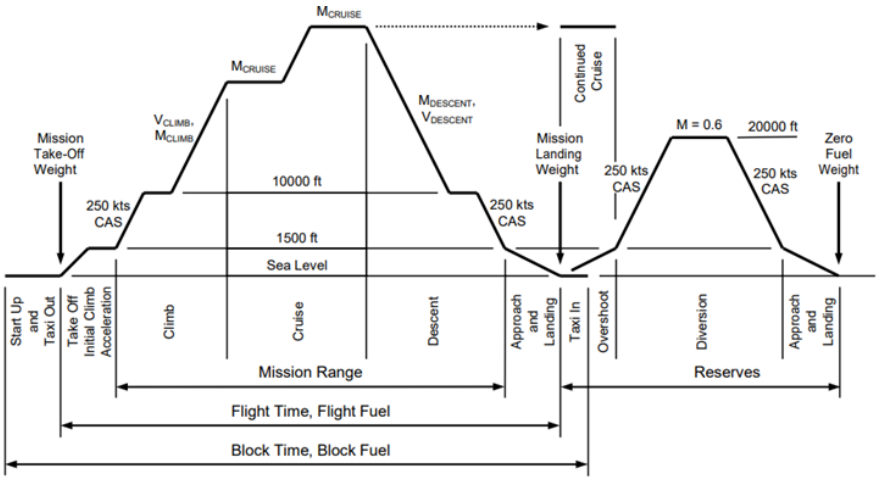


Figure 10.1 FAR Domestic Flight Profile.

Results:

500 nm Mission				
	Profile	Total Fuel / kg	Time/ mins	Distance/m
Mission / block fuel	Start-up, Taxi-Out	169	7	
	Take-off, initial Climb	448	11	
	Climb 1	274	4	27018
	Climb 2	823	14	177263
	Cruise	469	39	591193
	Descent 1	47	7	80271
	Descent 2	40	7	50255
	Approach, Landing	69	5	
	Taxi in	129	7	
Reserve	Continued Cruise	358		
	Overshoot	269		
	Diversion climb	783		
	Diversion cruise	120		
	Diversion Descent	69		
	SUM	4067	101	926000
	Mission Fuel Bur/ned	2468		

Table 10.1 Flight Profile Results

1000 nm Mission				
	Profile	Total Fuel / kg	Time/ mins	Distance/m
Mission / block fuel	Start-up, Taxi-Out	169	7	
	Take-off, initial Climb	447	11	
	Climb 1	274	4	27018
	Climb 2	823	14	177263
	Cruise	1204	100	1517193
	Descent 1	47	7	80271
	Descent 2	40	7	50255
	Approach, Landing	69	5	
	Taxi in	129	7	
Reserve	Continued Cruise	358		
	Overshoot	691		
	Diversion climb	782		
	Diversion cruise	120		
	Diversion Descent	69		
	SUM	5222	162	1852000
	Mission Fuel Burned	3202		

Table 10.2

10.2 Climb Performance

Our aircraft's climb is split into two regions of climb, 1500ft to 10000ft constant velocity climb of 250kts and a 10,000ft to 33,000ft constant Mach number climb of Mach number 0,65. Typically, high throttle settings are chosen during climb 80-100 %, this is due to the high frictional effects of low altitudes causing higher power requirements so limiting the time in these low altitudes is crucial. Climb phases burn large quantities of fuel and can be an uncomfortable phase of flight for customers, a hard climb time requirement of less than 25 mins is met by our plane with ours reaching cruise altitude in 19 minutes, the other hard requirement was keeping velocity under 250kts for altitudes less than 10,000ft which our first climb section achieves.

- Section 1 mean ROC: 12m/s
- Section 2 mean ROC: 11m/s

The two climb stages were calculated using the equations below, for the first stage velocity was constant and Excess power varied with altitude and for the second stage both velocity and excess power varied with altitude. Each section was split up into 100 minor sections, this enabled a more accurate calculation of how their respective Rate of Climbs varied with increasing altitude.

$$\text{Thrust} = \text{TA_altitude} \times \text{Throttle Setting}$$

$$\text{Excess Power} = (\text{Thrust-Drag}) \times V$$

$$\text{ROC} = \text{Excess Power} / W$$

Fuel consumption X_Distance travelled and Y_Distance travelled was calculated at each minor section and the sum of these values give the total for climb for each major climb section.

10.3 Cruise Performance

A similar loop is used to calculate the fuel burned and the distance travelled in climb was used for the cruise stage, however here altitude is constant, and the total distance is split into 100 minor sections with the total weight being recalculated after every minor section.

Key Cruise flight parameters of final design:

Mach Number	0.82
Altitude	31,000 ft
Thrust available	20832 N
SFC	0.05599 kg/hr/N
Thrust Required	12900 N

Table 10.3 Key Cruise Flight Parameters

10.4 Descent Performance

For an aircraft to descent Drag must exceed the thrust produced by the engines, during descent the thrust is switched to an idle setting. Bellow shows how the rate of descent varies with altitude for the two distinct sections 33000ft to 10000ft constant Mach number descent and 10000ft to 1500ft constant velocity section.

Similarly, to Rate of climb Rate of descent, fuel usage and was calculated using minor sections as altitude decreases and thrust available increases. Here however an idle throttle setting was used.

$$\text{Thrust} = \text{TA_altitude} \times \text{Throttle Setting}$$

$$\text{Excess Power} = (\text{Thrust-Drag}) \times V$$

$$\text{ROD} = -\text{Excess Power} / W$$

- Section 1 mean ROD: 10m/s
- Section 2 mean ROD: 7m/s

Fuel consumption X_Distance travelled and Y_Distance travelled was calculated at each minor section and the sum of these values give the total for climb for each major section.

10.5 Fuel Burned

- Fuel burned Is calculated using the equation bellow SFC varies with the plane's velocity and altitude and the time is the period of each minor calculated section.

$$\text{Fuel Burned} = \text{Thrust} \times \text{SFC} \times \text{Time}$$

10.6 Take-off and Landing Distances

Take-off Performance:

Take-off Distance Table / ISA at MTOW			
Altitude / ft	Ground Roll / m	clearance 35ft / m	Total Distance / m
0	1592	535	2127
1000	1670	546	2215
2000	1752	557	2309
3000	1839	569	2408
4000	1932	581	2513
5000	2030	594	2625
6000	2135	608	2743
7000	2247	622	2869
8000	2366	637	3003
8500	2428	645	3073

Table 10.4 Take-off Performance.

Take off has two distinct stages the ground roll and the clearance to 35ft flight. Our aircraft is reaching the requirement of Ancillary Operational Requirements as it can take-off at a variety of runways with different altitudes from sea level runways to runways at 8500ft.

The below equations were used to approximate the ground roll and clearance 35 ft x distance:

The average forces acting on the ground roll calculation was found by using 0.7 × V_lof, V_lof was taken as 144kts.

$$s_{LO} = \frac{1.44\,W^2}{g\,\rho_\infty S C_{L,\max}\{T - [D + \mu_r(W - L)]_{av}\}}$$

- h_c = clearance height (35 ft)
- YLof = climb angle
- ΔC_L = 0.1*CLmax

$$\text{Clearance 35 ft} = \frac{\frac{W}{S\ ylof}}{rho\ g\ \Delta C_L + \frac{h_c}{ylof}}$$

10.7 Landing Performance

- The Aircraft safety requirements of landing is our aircraft must be able to land within 1600m at an approach speed of 144kts our aircraft meets this requirement.

Results:

Landing Distance Table / ISA MLW	
Altitude / ft	Ground Roll / m
0	1150

Table 10.5 Landing Distance Table.

The below equation is used to approximate the ground roll of our aircraft:

$$s_L = \frac{1.69W^2}{g\rho_\infty SC_{L,\max}\{T_r + [D + \mu_r(W - L)]_{0.7V_r}\}}$$

10.8 One Engine inoperative Take-off performance

The below results are found for one Engine inoperative situation where our aircraft attempts to take-off, but one engine becomes inoperative after the ground roll. Calculation method used is the same as take-off distance calculation above however thrust halves after the ground roll.

One Engine Inoperative Take-off Distance Table / ISA MTOW			
Altitude / ft	Ground Roll / m	clearance 35ft / m	Total Distance / m
0	1592	1663	3255

Table 10.8 One Engine Inoperative Distance.

11. Aerodynamics

Aerodynamics are a crucial aspect of aircraft design that influences the overall performance, behaviour, and efficiency of an aircraft. This section explores the design choices made and their contribution to the aircraft's performance.

11.1 Aerofoil Selection

Choosing the optimum aerofoil is crucial to maximising the aircraft's aerodynamic performance. Factors such as: cruise Mach number, desired take-off & landing performance and total aircraft L/D are all deduced by this choice.

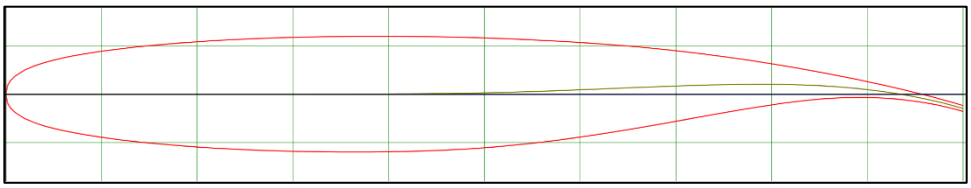


Figure 11.1 NASA 0712 Supercritical Aerofoil (Airfoil Tools,2023).

To maximise the lift during cruise and to reduce the effects of wave drag at our desired cruise speed we initially opted for a supercritical aerofoil. Specifically, we chose the NASA SC(2) - 0712 aerofoil due to the availability of existing research into this series of supercritical aerofoils.

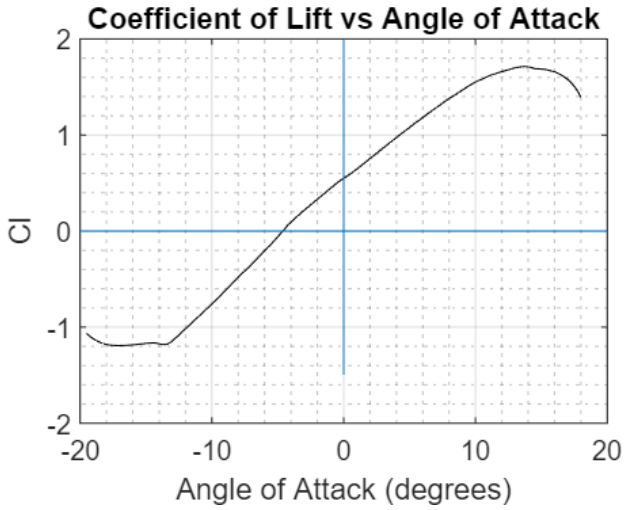


Figure 11.2 NASA 0712 Coefficient of Lift vs Angle of Attack.

The behaviour of the aerofoil can be seen in Figure (11.1.). It was an aerofoil that was particularly desirable due to its relatively high coefficient of lift at low angle of attack (AoA). Additionally, the maximum coefficient of lift occurs at a high AoA denoting good stall characteristics. The rounded peak of the graph implies a benign stall which can make stall recovery easier. This is favourable compared to a sharp peak which signifies that at stall there is a sudden and very severe decrease in lift leading to a more difficult stall recovery.

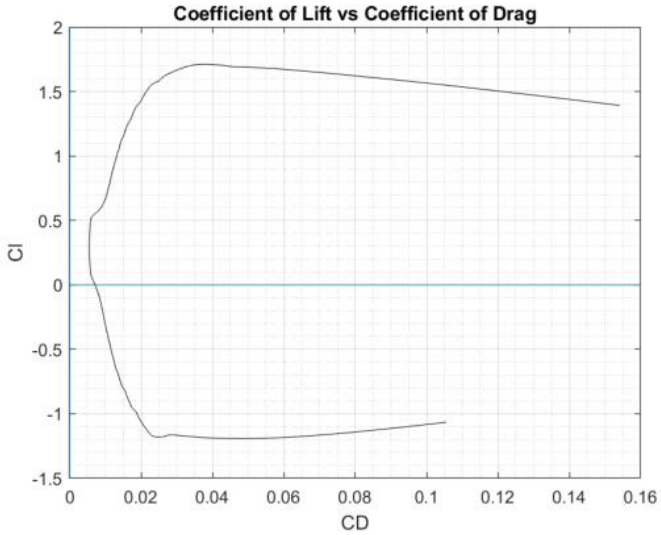


Figure 11.3 Coefficient of Lift vs Coefficient of Drag

Figure (11.2) displays the relationship between the coefficient of lift and the coefficient of drag for the NASA 0712. Crucially, the lowest coefficient of drag occurs in the region when $0 < Cl < 0.51$, this is close to the design coefficient of lift and implies that phases of flight in this Cl range will be the most efficient. At the maximum coefficient of lift the coefficient of drag is still relatively low, once again this is beneficial for optimum climb efficiency.

11.2 Critical Mach Number

Cruise speed is a vital consideration when developing a competitive short haul aircraft. Shorter flight times increase customer satisfaction whilst maximising airline revenue by offering more frequent flights. As such, market analysis indicates that cruise speeds vary between Mach 0.78 & Mach 0.85 for similar turbofan aircraft. Whilst flying faster may improve aircraft utilisation, there is a trade-off between journey time and efficiency. At higher Mach numbers, localised regions of the aircraft may develop supersonic flow which leads to shockwaves and associated wave drag. To mitigate the effects of wave drag, the aircraft maximum cruise speed will set below the critical Mach number. This also has the benefit of reducing the structural strength of the airframe since external forces are less at subsonic speeds. Due to the aircraft's large wingspan, it was an important design consideration that external forces due to a high cruise speed did not induce substantial flutter effects. Flutter effects can seriously jeopardise the structural integrity of the aircraft. As such, the decided cruise speed would be significantly lower than the critical Mach number.

The methodology to obtain the Critical Mach Number (M_{Cr}) was taken from Anderson (2016, Chp, 11.6). Using the equations below, a value was obtained.

$$C_{p,cr} = \frac{2}{\gamma {M_{cr}}^2}\left[\left(\frac{1+[(\gamma-1)/2]{M_{cr}}^2}{1+(\gamma-1)/2}\right)^{\gamma/(\gamma-1)} - 1\right]$$

$$C_p = \frac{C_{p,0}}{\sqrt{1-M^2}}$$

M in these equations represents the range of Mach numbers (0.5 – 1.25) used to obtain a figure below. The point at which the two equations meet allows the M_{Cr} to be obtained. This returned a value of Mach 0.89.

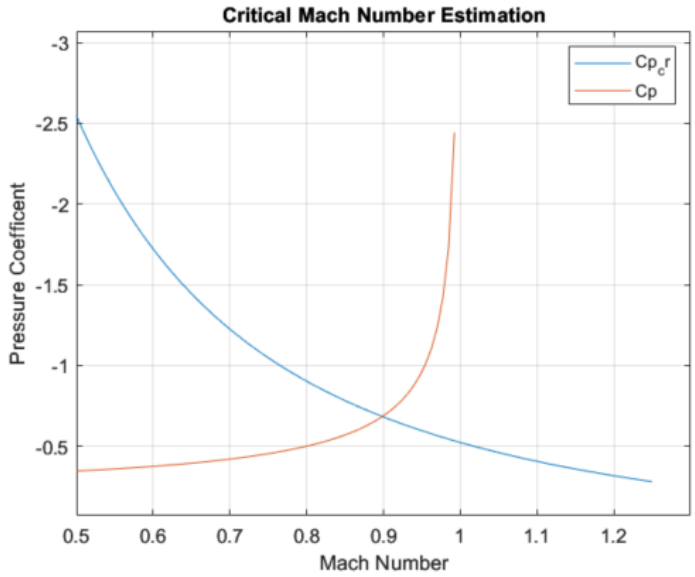


Figure 11.4 Critical Mach Number Estimation.

11.3 Design Lift Coefficient

The coefficient of lift during cruise can be calculated from the requirement of the lift produced being equal to the weight of the aircraft during steady level flight.

$$C_{L_{cruise}} = \frac{2L}{\rho S V^2} = \frac{2W}{\rho S V^2}$$

For the predetermined cruise altitude and cruise speed, a design lift coefficient of 0.5304 was calculated. The weight of the aircraft was taken from an average of the post-climb cruise weight to the pre-decent cruise weight.

11.4 Drag Build Up

To achieve the desired project goal of maximising efficiency it was paramount that drag be reduced to a minimum. Detailed below are the constituent drag elements that sum to find the overall coefficient of drag for the aircraft.

$$C_{Di} = \frac{C_L^2}{\pi A R e}$$

Induced drag (C_{Di}) is primarily dependent on the wing's lift generation and aspect ratio. Typically, high C_L values are preferred so reducing this term is unfavourable. Instead, increasing the aspect ratio stands to be the best way of reducing induced drag penalties whilst maintaining lift generating capabilities. The projects final design iteration settled upon an aspect ratio of 20, this led to C_{Di} value of 0.005597.

The following equations detail the process that was used to calculate the profile drag of each of the aircraft's main components. The example below demonstrates the methodology used for calculating the fuselage profile drag.

$$C_{D0} = C_f FQ \left[\frac{Swet}{Sref}\right]$$

$$C_f = [0.455]/[(\log Re)^{2.58}(1 + 0.144M^2)^{0.65}]$$

$$F = 1 + 2.2/(\lambda)^{1.5} - 0.9/(\lambda)^{3.0}$$

$$Q = 1.0$$

Component	CD0 Value
Fuselage	0.002029
Wing	0.004277
Strut	0.004829
Vertical Stabiliser	0.004206
Horizontal Stabiliser	0.004609
Nacelle	0.004840
Component Total	**0.02479**

Table 11.1

The process is replicated for each component detailed in Table 11.1 Values of the coefficient of friction (Cf), Form Factor (F), and Interference Factor (Q) vary due to the difference in geometry. This profile drag methodology was taken from "Civil Jet Aircraft Design" (Jenkinson,1999). Recommended are appropriate interference factors ranging from 1 – 1.5, while each component has a prescribed form factor equation. Due to every component's Reynolds number exceeding the laminar boundary layer condition (Re ≤ 500,000) , the turbulent boundary layer C_f equation was used.

In "Aircraft Design: A Conceptual Approach" (Raymer, 1989), drag effects from leakage and miscellaneous effects are considered to be roughly 3% of $C_{D0Total}$ & 0.0021 respectively. Wave drag effects were ignored due to the aircraft's max cruise speed being below M_{Cr}.

$$C_D = \sum C_{D0} + C_{Di}$$

The final C_D of the aircraft in its cruise configuration totalled 0.0331, with this value gathered a final cruise L/D of 16.03 was calculated. In comparison, "today's airliners achieve an *L/D* over 20 at Mach 0.85" [28]. While our aircraft has a lower L/D this was achieved without using significant optimisation techniques in addition to the assumption that the flow all over the aircraft was entirely turbulent.

11.5 High Lift Devices

To increase the versatility and competitiveness of the aircraft, high lift devices will be installed that will allow greater control and performance in certain conditions. The key purpose will be to minimise both take-off and landing distances through increased lift generation. The capability to land and take-off at shorter runways would give this aircraft a competitive edge. Fowler flaps have been selected for the aircraft due to their two-stage function. The first being extension that can greatly increase lift at the cost of a minimal increase in drag (ideal for take-off). The second is a downward movement greatly increasing drag whilst still generating high lift (ideal for landing).

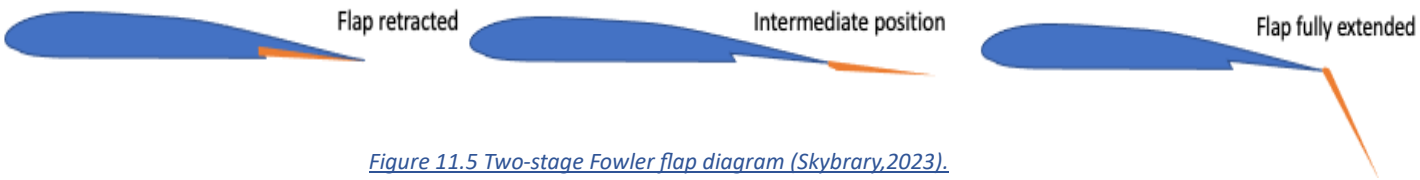


Figure 11.5 Two-stage Fowler flap diagram (Skybrary,2023).

During take-off, a 10-degree flap deflection was chosen to increase lift whilst allowing a marginal increase in drag. In landing, where large amounts of lift are required and high drag can be tolerated to shave off approach speed, a 20-degree flap deflection angle was chosen. Due to the increase in surface area, there is a considerable effect on the aircrafts total coefficient of drag. This is reflected in the aircrafts take-off and landing lift to drag ratios. These C_D values also reflect the increase due to landing gear drag effects.

Deflection Angle	**10° (Take-off)**	**20° (Landing)**
C_{Lmax}	1.8220	2.0740
C_D	0.1020	0.1232
L/D	17.86	16.83

Table 11.2 Deflection Angle at Take-off and Landing.

11.6 Strut Aerodynamics

An essential element of this project's innovative aircraft configuration was the integration of a strut. While the strut's primary role is to support the wing's weight over its larger span, its aerodynamic characteristics cannot be ignored. Due to the limitations of the report, no computational fluid dynamics (CFD) modelling could be conducted so many of the strut related design choices are influenced from research. An important consideration from research suggests that "aircraft with a higher Mach number (Ma = 0.8) may benefit from loading the strut due to compressibility effects" [29]. Since the cruise Mach of the proposed aircraft is in excess of this, it must be considered. Furthermore, another study implies an "unstacked geometry of the inboard strut and inboard wing" [30] tie-in locations prevent flow separation and as a result compressibility drag can be reduced. These suggestions may however have to be explored further later with CFD analysis to properly justify their effects.

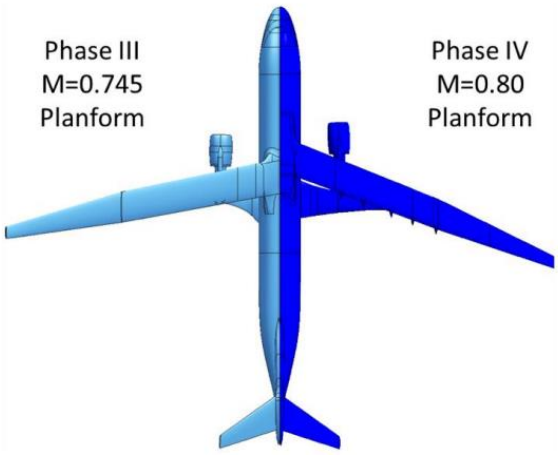


Figure 11.6 (Harrison et al., 2023).

Figure (11.6) demonstrates the "unstacking geometry" feature between the Phase III and Phase IV SUGAR Truss Braced aircraft. Notice the geometry differences resulting from Mach number increase.

11.7 Aerodynamic Performance Iterations

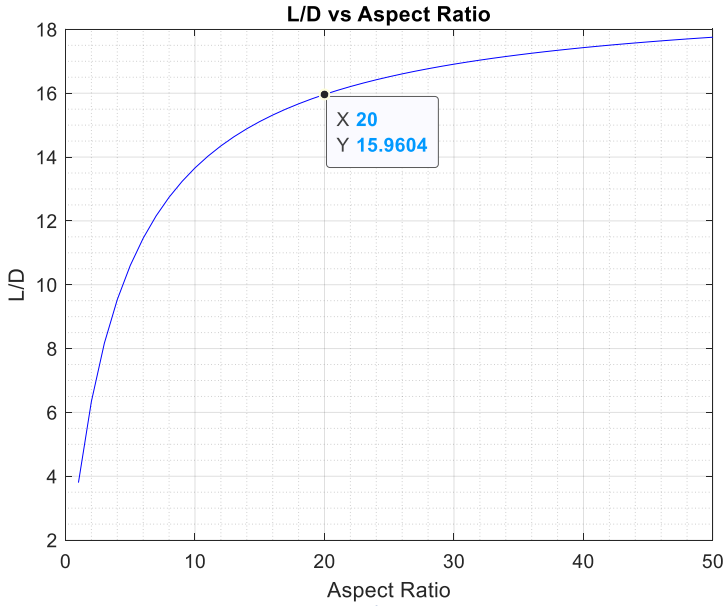


Figure 11.7 L/D vs Aspect Ratio

Figure (11.7) implies that greater aspect ratios increase the total lift to drag ratio (L/D). Optimising for efficiency was a key goal of this study so a higher L/D ratio was preferred. An aspect ratio of 20 was settled on as it provided a compromise between performance and the limitations of what is possible structurally. Additionally, previous research indicates that aspect ratios between 18-21 are optimal for strut-braced wing configurations.

Performance iterations

Our cruise Mach number went through three design iterations- at first a cruise speed of 0.75 was chosen, this was due to comparisons between similar transonic airliners. Then considering our critical Mach number of 0.89 we know the cruise Mach number can safely be increased to a point which is safely below the critical, this is important for our design as Mach effects can cause high moments which can potentially cause wing flutter in our design due to the higher span than conventual aircraft; to mitigate this a Mach number of 0.82 was chosen.

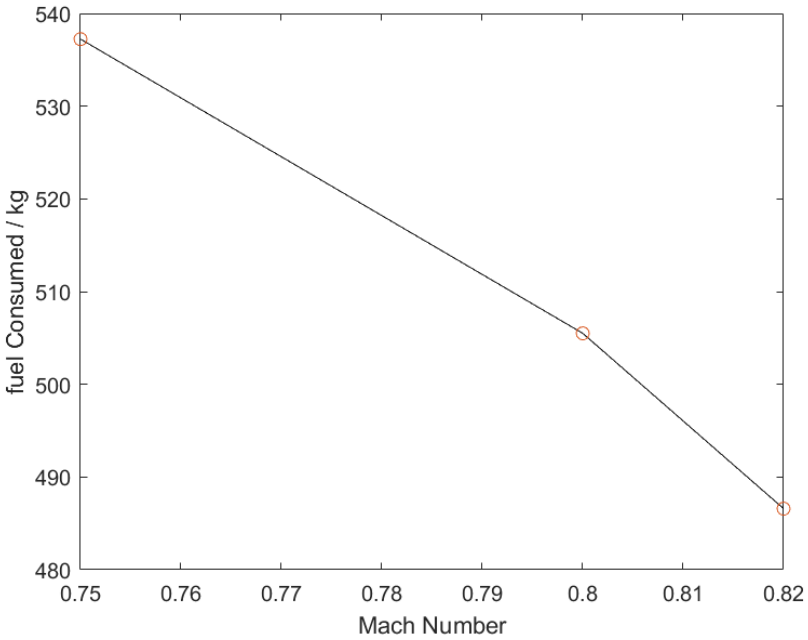


Figure 11.8 fuel consumed at cruise against cruise Mach number (500

Design cruise altitude went through a further 5 design iterations, an altitude of 31000ft was chosen, this altitude provides adequate fuel reduction as well as providing various other advantages. For

example, con trail reduction which is suggested as a result of lower altitudes and improved passenger comfort as cruise providing consistent flight path angle and steadiness compared to climb and descent phase as well as lower altitudes reducing the risk of hypoxia (sickness related to decompression/ pressure loss). This reduction in fuel consumption is a result of the air density effecting the drag and power requirements and the specific fuel consumption of our engine which at our cruise Mach number of Mach 0.82 is lowest at 31,000 ft figure 11.9.

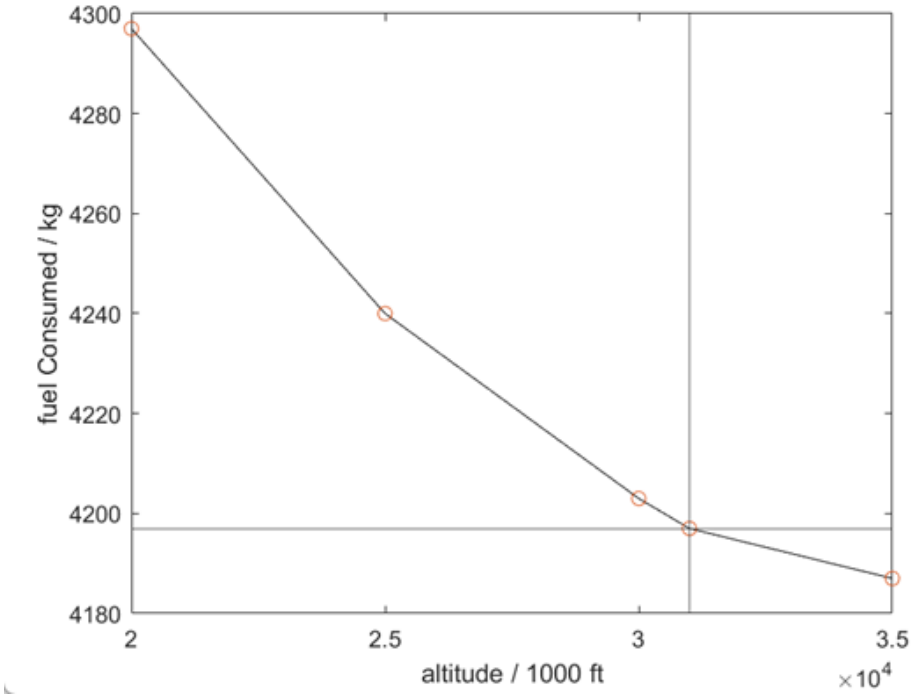


Figure 11.9 fuel consumed against altitude (500 nm flight)

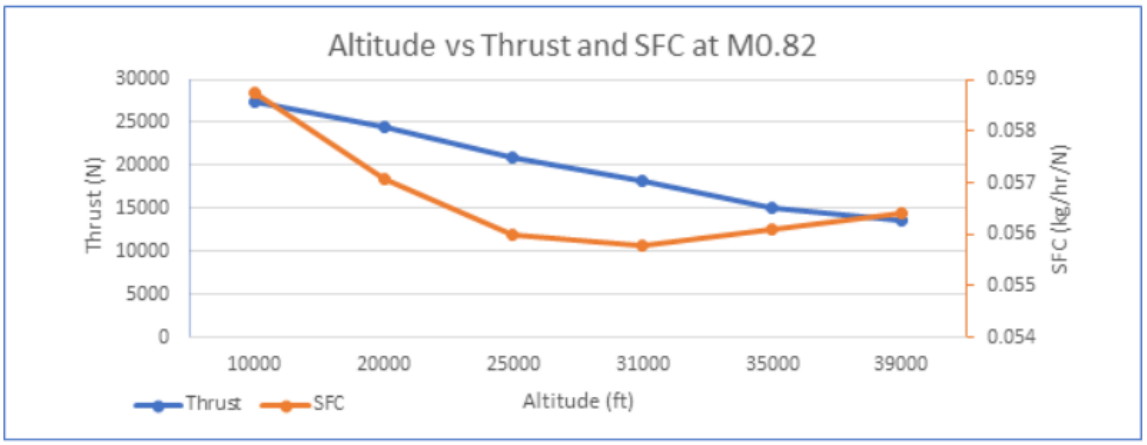


Figure 11.10 Specific consumption against altitude of engines.

12. Stability & Control

12.1 Empennage Justification

The tail configuration chosen for the aircraft under study is a T-Tail. This geometry was chosen as it removes the tail from the exhaust blast created by the high wing mounted engines. For transonic aircraft a T-tail configuration may improve pitch control effectiveness because the elevator is not in disturbed air behind the fuselage, particularly at moderate angles of attack.

12. 2 Static Longitudinal Stability Specifications

The Horizontal tail should be large enough and the elevator powerful enough to enable the pilot to rotate the aircraft the take-off run to the required angle of attack. This condition is termed as the nose wheel lift-off condition.

12. 3 Static longitudinal stability analysis to estimate a revised horizontal tail area

The proposed method by Tulapurkara, E.G. (2007) for the revised horizontal and vertical tail areas is employed below. The method draws from other literature and is referenced throughout the process.

Equation symbols and definitions are provided in the nomenclature section of the report.

Chapter 16 of Raymers approach, reference 3, presents a curve for $(C_{ma})_{stick\text{-}free}$ vs M for different aircraft. From this curve the appropriate value of $C_{m\alpha}$ is obtained. From this value of $C_{m\alpha}$, the static margin stick-free is calculated.

The contribution of wing, fuselage, power, and horizontal tail are worked out for cruise flight condition with aircraft weight equal to the design gross weight. The method below gives the required value of tail volume ratio V_H. Subsequently the area of horizontal tail can then be calculated.

The horizontal tail area can be calculated as follows:

$$\frac{dC_m}{dC_L} = \left(\frac{dC_m}{dC_L}\right)_{wing} + \left(\frac{dC_m}{dC_L}\right)_{fuselage} + \left(\frac{dC_m}{dC_L}\right)_{nacelle} + \left(\frac{dC_m}{dC_L}\right)_{power} + \left(\frac{dC_m}{dC_L}\right)_{h.tail} \quad [1]$$

Contribution of the fuselage $\left(\frac{dC_m}{dC_L}\right)_{fuselage}$

From chapter 16 in the Raymer, reference 3, approach to aircraft design an approximate expression for this contribution is:

$$\frac{dC_m}{dC_L} = \frac{Kf\ W_f\ l_f}{S\ c\ a_w} \quad [2]$$ The value of Kf is obtained as 0.0119 from Fig.16.14 of Ref.4.

The value obtained for the contribution of the fuselage was 0.1958.

The contribution of the nacelle to $\left(\frac{dC_m}{dC_L}\right)$ neglected during this stage.

From reference 5, chapter 5 a simplified contribution of power can be obtained.

$\frac{dC_m}{dC_L} = \frac{Tt_p}{Wc}$ [3] Hence a value obtained for the contribution of power at cruise was 0.0045.

The equation of horizontal tail in stick free case is , $\left(\frac{dC_m}{dC_L}\right)_{h.tail}$ and is obtained from reference 6, chapter 13. The equation is as follows:

$\frac{dC_m}{dC_L} = -\frac{a_t}{a_w} V_H \, n_t \left(1 - \frac{de}{da}\right)\left(1\tau \frac{C_{hat}}{Choe}\right)$ [4] From this a value of 0.274 was obtained.

Note that dε / dα is the downwash and is estimated using the following approximate formula.

$\frac{de}{da} = \frac{114.6 * a_w}{\pi A}$ = 0.43

From chapter 6 of reference 3, the recommended value of C_{ma}, for a jet transport flying at a Mach no. 0.82 is -1.15.

Hence, $\frac{dC_m}{dC_L} = \frac{C_{ma}}{a_w}$ [6] = -0.1702, If we sub this value into equation [1] = 1.3501

Using the following equation, a horizontal tail area can be obtained.

Rearrange and subbing into equation [1] the horizontal volume tail ratio V_H can then be found.

$S_h = \frac{V_h S b}{l_h}$ [7] A value of 27.5 m^2 is obtained.

12. 4 Lateral Stability Specifications

The directional static stability criterion, dC_n / dβ, should be positive for any flight speed greater than 1.2 times the stalling speed as specified by FAR 25.149.

The yawing moment control (rudder) must be powerful enough to counteract the yawing moment encountered (a) in roll (adverse yaw), (b) in crosswind landing or take-off, (c) when one engine is inoperative for multi-engine airplanes. The spin recovery is also affected primarily by the rudder.

The equation for directional stability is derived from reference 6, chapter 5.

$\frac{dC_n}{d\beta} = C_{n\beta_{wing}} + C_{n\beta_{fuselage}} + C_{n\beta_{power}} + C_{n\beta_{V.tail}}$ [8]

12. 5 Lateral Stability analysis to estimate a revised vertical tail area

In this preliminary analysis of directional static stability, the contributions of wing, power and interference effects are ignored. It is further assumed that the contributions to $C_{n\beta}$ due to wing sweep and low wing position cancel each other.

An approximate formula from reference 7 chapter 1-9 is:

$(C_{n\beta})_{fuselage} = \frac{-k_n V_n}{28.7Sb}$ Hence the value for fuselage contribution is -0.0014 [9]

$(C_{n\beta})_{V.tail} = a_v \frac{S_v}{S}\frac{l_v}{b}$ as $V_v = \frac{S_v}{S}\frac{l_v}{b}$ hence an expression of 0.0378 * Vv was obtained. Where V_v is the volume ratio of the vertical tail. [10] [11]

The value for $C_{n\beta(desirable)}$ is given by reference 5, chapter 8:

$C_{n\beta(desirable)} = 0.0005\left[\frac{W}{b^2}\right]^{-0.5}$ [12] And a value of 0.0015 is found.

Using equation [7] and substituting values 0.0015 = -0.0014 + 0.0378 * Vv Or Vv = 0.0765

Hence the vertical tail area can be found:

$S_v = \frac{V_v Sb}{l_v}$ = 31.9 m^2 n [13]

12. 6 Control surface sizing

The following values are chosen based on three-view drawings of similar aircraft and data collection.

S_{ele}/S_h = 0.22 and S_{rud}/S_v = 0.22

Therefore, the control surface areas are:

S_{ele} = 27.5 * 0.22 = 6.05 m^2 and S_{rud} = 32.9 * 0.22 = 7.2 m^2

Trailing edge flap type: Fowler flaps

Leading edge high

S_{flap}/S = 0.17

S_{slat}/S = 0.10

B_{flap}/b = 0.74

Hence, the area of the various flaps is:

Area of T.E flaps = 112.6 * 0.17 = 19.142 m^2

Area of L.E slats = 112.6 * 0.1 = 11.26 m^2

Bflap (span) = (47.45-5.73) * 0.74 = 30.87 m, note that here the distance 5.73 m represents the span of the folding wing tips.

From the calculation S_h and S_v, we are then able to use the aspect ratios of the relative components to calculate the horizontal and vertical tail spans, with the aspect ratios of spans being, 5 & 1.82 (ref) respectively, both these ARs are based on the A320's design. The formula below can then be rearranged used to calculate the spans of the tails. x is this case represents either the horizontal or vertical components.

$$AR_x = {b_x}^2 \Big/ S_x$$

Hence, the values are:

$b_{htail} = 11.74m$

$b_{vtail} = 12.64m$

The root and tip chords can then be found as:

$c_{rh} = 0.6785m, c_{th} = 0.1737m$

$c_{rv} = 0.6072m, c_{tv} = 0.1840m$

12. 7 Actuator Sizing

Hinge moments are needed in the design of aircraft, as they determine the forces required to move the control surfaces such as elevators and rudders. The coefficient hinge moment of the elevator can be assumed to be linear. C_{he}, can be derived by the calculating b_1 & b_2, with b_3 can be assumed to be zero for tab-less flight. b_2 was calculated at a 20-degree deflection. Using formulas from Etikin, B (1996)[32]:

$$b_1 = \frac{\partial C_{he}}{\partial \alpha_t}, b_2 = \frac{\partial C_{he}}{\partial \delta_e}, b_3 = \frac{\partial C_{he}}{\partial \delta_t}$$

$b_1 = -0.2338\,, b_2 = -1.44224$

From there, the tail angle of attack was calculated, allowing the coefficient hinge moment for trimmed flight to be calculated using the following:

$$a_t = a_{wb} - \left(\varepsilon_0 + \frac{\partial \varepsilon}{\partial a} a_{wb}\right) - i_t = -0.1002\, rad$$

$C_{m_e} = b_0 + b_1 a_t + b_2 \delta_t$ = 0.00069201

Using the dynamic pressure, MAC of the elevator, tail efficiency and the coefficient hinge moment, the total hinge moment for trim can be calculated, allowing use to calculate the stick force

$H_e = n\, q\, C_{mac_e} C_{m_e} = 2.0324\, n.m$

$P_e = GH_e = 8.1297$ N

The reason for a low gearing ratio, is to reduce the relative force need to be applied to the stick to result in a larger movement of the elevator. This therefore means that pilots can make finer more accurate adjustments with less physical strain.

A similar process can be applied to the rudder to calculate the pedal force required to alter the rudder to counter any cross winds, where beta and sigma represent sideslip and sidewash receptively:

$a_f = -\beta + \sigma$

$C_{h_r} = b_1 a_f - b_2 \delta_r = 7.2576$

$H_r = nq C_{mac_r} C_{m_r} = 72.6722$

Finally, the pedal rudder force can be calculated by:

$P_r = GH_r = 290.6886N$

Hinge moments allow the final sizing of the actuators, from the area of the rods to volumetric displacement of the rods, other factors such as torque can be calculated using the know force of the stick and multiplying by the distance from the stick position to the control surface. This allows the power required of the motor to move the corresponding components.

There are different types of actuators including hydraulic, electric, and electro-hydraulic. For the design, electro-hydraulic actuators have been chosen, due to the preferable characteristics, such as: compact and lightweight, low-maintenance, energy-efficient and most of all high degree of precision. Alongside a small stick force further allows the pilot to perform more accurate and acute changes allowing for a smoother and more reliable flight.

12. 8 Evaluation of a Fly-by-Wire Control System

The fly-by-wire system is a semi-automatic computer system, which controls converts the pilots flight controls and movements into an electronic signal. This is then interpreted by the flight control computers resulting in adjustments in high lift devices.

Other methods of flight control systems include, fly-by-cable and hydraulic systems, these systems give the pilot a more direct hands on feel when adjusting the actuators, however due to the number of mechanical parts, these systems need constant monitoring and maintenance, increasing costs and potentially layover time. As well as this, the increase in mechanical parts result in an increase in weight, as well as this, bulky parts such as levers and pulleys need more room. The use of electronics eliminates these variables, as well as provides a faster input to response time.

As previously mentioned, the benefits of a heightened response time allow the aircraft pilots to obtain smoother flying, as well as safer flying as the aircraft's computers are programmed to maintain the aircraft in the limits of its' flight envelope. If the pilot fails to adjust the controls to react to a situation the system will automatically adjust the plane accordingly.

The aircraft will have the fly-by-wire system designed by Thales; this supplies the full flight control system. This is predominantly used within the multiple Airbus aircraft families, as well as other aircraft manufacturers such as Bombardier and Global. This system is also certified by the EASA and by other aviation agencies.

13. Structures

13.1 Shear Force & Bending Moment Diagrams

To get an understanding of the loads effecting the wings of the aircraft, the bending moments as well as the shear forces were analysed to best understand how to construct the structure of the plane. It was found that forces were concentrated around the root of the wing due to the bending moments acting along the span. Also, most of the lift is created at the root of the wing due to the larger chord length and thickness.

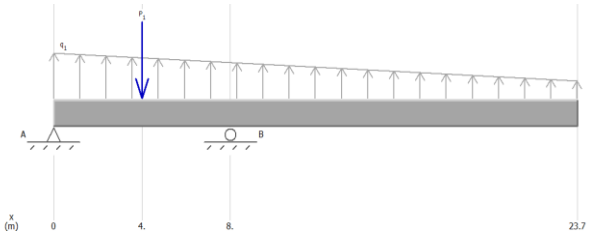


Figure 13.1 Simplified beam diagram with strut at 8m.

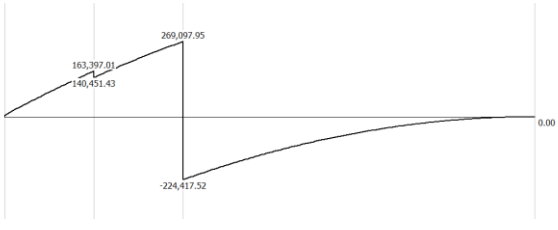


Figure 13.2 Shear force diagram with strut at 8m.

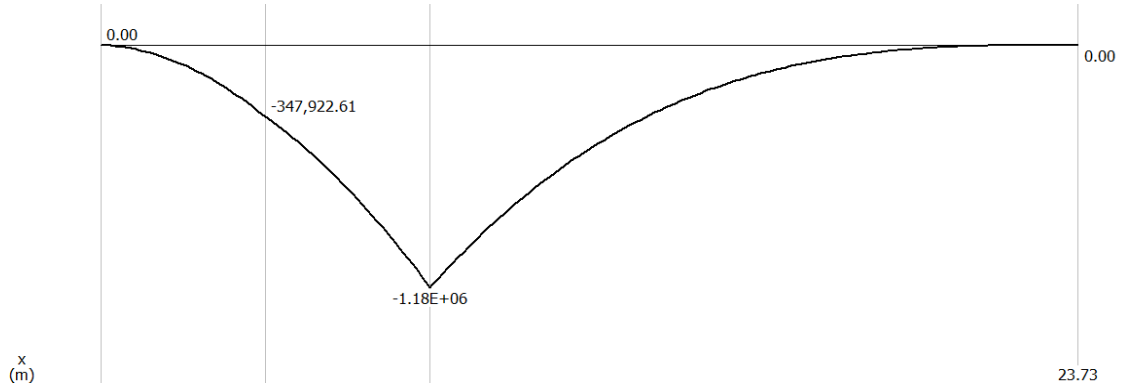


Figure 13.3 Bending moment diagram with strut at 8m

To understand if the wing was able to resist all forces during flight, we designed a 2.5g test case to simulate the wing loading during a co-ordinated turn. To do this, we simplified the wing to beam, assuming the engine as a point load. The distributed weight and lift were also considered by applying them as triangular loads. Reaction A represents the wing-fuselage junction, whilst Reaction B represents the strut.

The lift can be seen to reduce spanwise along the wing. This is due to the trapezoidal shape, which tapers, meaning it gets smaller. This means that lift does not have an equal distribution along the wing, meaning that we needed to calculate this distribution, using:

$$L(y) = \frac{2L}{b(1+\lambda)}\left[1 - \frac{2y}{b}(1-\lambda)\right]$$

Weight also reduces with span, using:

$$W(y) = \frac{2W}{b(1+\lambda)}\left[1 - \frac{2y}{b}(1-\lambda)\right]$$

By taking the difference between these equations we can find the net load on the wing used in the bending moment diagram.

13.2 Strut Analysis

It can be seen in the above diagrams that having the strut initially at 8m along the span resulted in some unwanted spikes in shear and bending forces. To distribute the forces along the wing evenly, we decided place the strut at different lengths along the span to see what results we could achieve. As a result of this, we found out that 15.5m was the optimum placement for the strut to ensure that the strut wasn't taking too much weight. By having the strut placement at 15m rather than 8, we were able to equalise the reaction forces between the root junction and the strut.

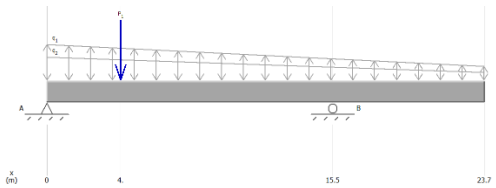
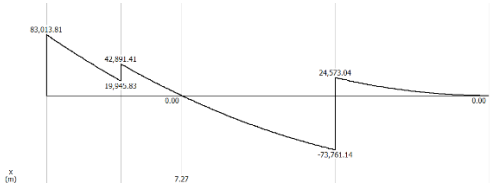


Figure 13.4 Simplified beam diagram with strut at 15m



13.5 Shear force diagram with strut at 15m.

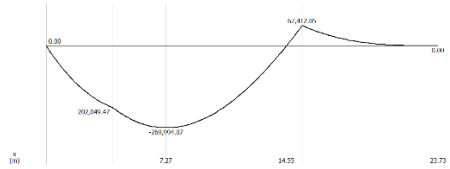


Figure 13.6 Bending moment diagram with strut at 15m

13.3 Wing Box Design

During this project, the design has been optimized for having a larger aspect ratio to reduce induced drag and have a higher lift to drag ratio. However, in doing so we have created some setbacks, such as having larger bending and shear forces through the wing structure. To reduces the effect of this, we have added a strut to relieve some of the forces acting on the wing root. This will prevent the wing from shearing off as well as any buckling that may occur, should the wing be unsupported.

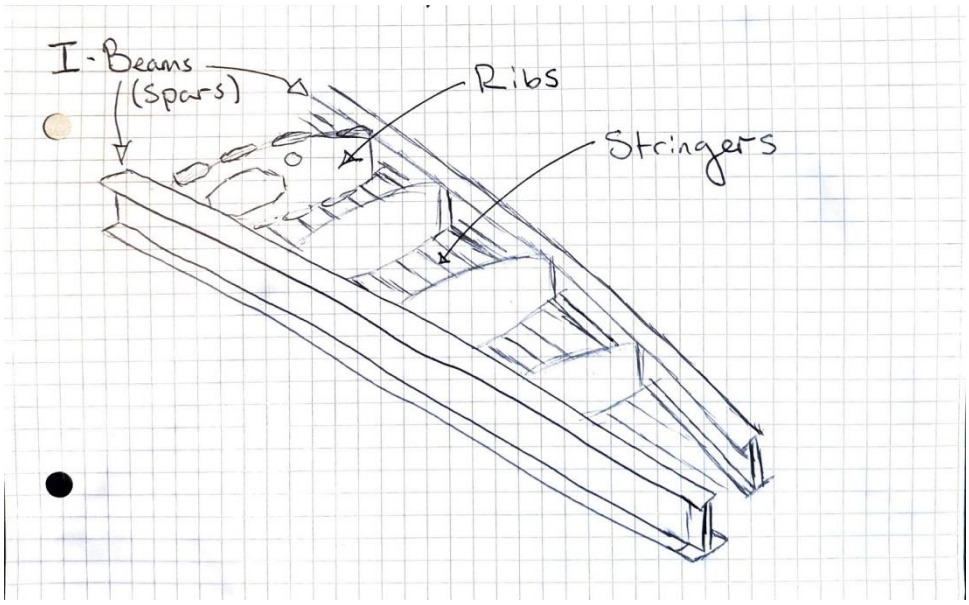


Figure 13.7 Sketch of a wing box.

To withstand loading during flight, the wing will be constructed using stiffeners such as Ribs, Spars, and stringers. Stringers are used to supply stiffness along the span of the skin panels. Ribs are used to form the skin panel into its aerodynamic shape, whilst the spars are there to supply the main support for the structure and support the ribs.

13.3.1 Spars

The wing box design shall include a Leading-Edge Spar and a Trailing Edge Spar to give support across the wing. The Spars will essentially be I-beams due to the higher value of Second Moment of area, meaning there will be less displacement when a load is applied meaning the wings will not flex as much compared to a different shaped beam.

13.3.2 Ribs

Along the span of the wing, there will be 38 ribs along the 23.73m half-span. This equals a rib every 0.62m to ensure the aerofoil profile is kept across the span of the wing.

13.3.3 Stringers

There will be 17 rows of stringers which will reduces as a row reaches the edge of the geometry of the skin panels. This means there will be more at the root of the wing allowing for more stiffness where there will be more shear forces acting on the wing.

13.3.4 Skin Distribution

Along the span of the wing, the thickness of the skin panels is not constant. This is due to the forces along the wing not being constant. The skin will be thicker where loads are more concentrated, such as the engine mounting pylon, the wing root, and the strut-wing junction. This will allow for less strain about these locations.

13.4 Fuselage Structure Design

The fuselage will be constructed using a series of ring frames, as well as stringers on the fuselage surface and longerons joining the ring frames together for support.

The fuselage has a diameter of 3.77m and therefore a circumference of 11.84m, for the purpose of structural integrity it was decided there would be 70 stringers to have 1 stringer per 0.17m. The fuselage is also 38.7m in length again for structural purposes it was decided there would be 68 rings along its length to have a ring every 0.57m. These numbers were mostly based off estimations from existing aircraft such as the Boeing 787 and airbus a320 which had roughly 1 stringer per 1.74m. In terms of the rings, since each window is about 0.52 meters in width, the rings had to be further apart then that. This maximizes the aircrafts anti buckling capabilities which is desired due to aircraft skin only being a few millimetres thick. The diagrams below show what the force distribution across the fuselage should look like in a 2.5g scenario and the affects that has along the fuselage in terms of bending moments and sheer forces.

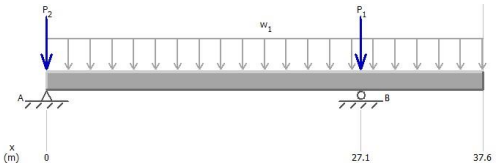


Figure 13.8 - Simplified beam diagram of the fuselage.

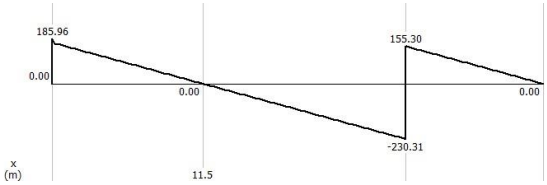


Figure 13.9 Shear force diagram.

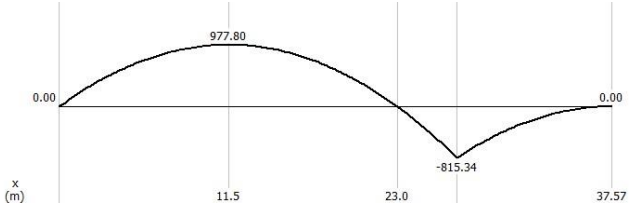


Figure 13.10 - Bending moment diagram.

The Fuselage will be pressurised to 8PSI which equates to 55158.1Pa. The atmospheric pressure at a 33,000ft cruising altitude is roughly around 25,000Pa. This means that the forces acting upon the fuselage will be 30,158.1Pa. As previously mentioned, stingers will be used to stiffen the fuselage skin. More specifically, “Hat” stringers will be used, due to their ability to also be used as piping for gases. For example, pressure relief.

13.5 Fuselage / Wing Root Junction

Usually, the wing is not attached to the fuselage, rather the fuselage is built around the wing. To do this, there is a central wing box found where the wing fuselage junction will be. For maximum structural integrity, the wings are joined to the spar of the central wing box, as well as other fasteners which join the upper and lower skin panels. Thus, once joined the wing is one continuous object that runs through the fuselage.

The wing box for this aircraft shall be located on top of the fuselage due to the high wing configuration. The central main wing box shall be incorporated into the fuselage for maximum aerodynamic efficiency. The central wing box will fasten to the fuselage using multiple fasteners to ensure a strong connection. The central wing box shall be constructed like the wings using to spars as well as other ribs for structural integrity instead of aerodynamic shape.

The wings will then slot into the central wing box. The spars will be bolted together to form one continuous spar, and the skin panel shall be attached using numerous fasteners to ensure a strong connection.

Aircraft sealant will be used between mating faces as well as over the top of bolt tails and head to provide a fuel tight seal where necessary. Connections between surfaces shall be made using lightning cables to ensure that all components are sufficiently earthed. This way electricity will not arc between surfaces causing fuel fumes to ignite.

13.6 Wing Tip Mechanics

Retractable wing tips were implemented on our aircraft to increase aerodynamic efficiency by increasing wingspan, reducing the strength of wing tip vortices, decreasing induced drag, and therefore increasing fuel efficiency. A hydraulic system will be used to control the wing tips. This involves a hydraulic power pack (an electric motor driving a small pump) which then controls hydraulic actuator allowing extension and retraction. Metal seals will be used to avoid leaks in the system.

13.7 Materials and Manufacturing

T300 Standard Modulus Carbon Fibre

FIBER PROPERTIES

PROPERTY	ENGLISH	METRIC	METHOD
Tensile Strength	512 ksi	3,530 MPa	TY-030B-01
Tensile Modulus	33.4 Msi	230 GPa	TY-030B-01
Strain at Failure		1.5%	TY-030B-01
Density		1.76 g/cm^3	TY-030B-02

COMPOSITE PROPERTIES

PROPERTY	ENGLISH	METRIC	METHOD
Tensile Strength*	264 ksi	1,820 MPa	ASTM D-3039
Tensile Modulus*	20 Msi	140 GPa	ASTM D-3039
Tensile Strain		1.26%	ASTM D-3039
Compressive Strength*	213 ksi	1,470 MPa	SACMASRM1R-94
Flexural Strength*	260 ksi	1,790 MPa	ASTM D-790
Flexural Modulus*	18 Msi	123 GPa	ASTM D-790
ILSS	14 ksi	94.1 MPa	SACMASRM1R-94
In Plain Shear Strength	14 ksi	95 MPa	ASTM D-3518
90° Tensile Strength	11 ksi	76 MPa	ASTM D-3039

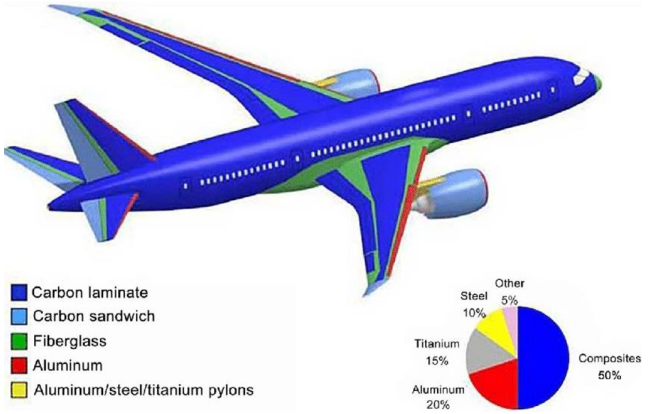


Figure 13.11

Aircraft material distribution based off Boeing 787 and all materials, and their properties sourced from companies known for supplying their materials to the aerospace industry for aircraft use.

Some parts of the aircraft such as the wing root and nose may need stronger carbon fiber then the T300. However, the TORAYCA company produces much stronger variants like the T1100G or M46J which could be used as an alternative.

Aluminium 7475-T7651 Properties

Tensile Strength	462 MPa
Tensile Modulus	71.7 GPa
Elongation at Break	12%
Density	2.81 g/cm^3

Titanium Ti-8Al-1Mo-1V Properties

Tensile Strength	1070 MPa
Tensile Modulus	120 GPa
Elongation at Break	17%
Density	4.37 g/cm^3

AISI Type S21904 Stainless Steel Properties

Tensile Strength	538 MPa
Tensile Modulus	200 GPa
Elongation at Break	42%
Density	7.83 g/cm^3

13.7 Material Choice in Aircraft

Primary aircraft structure and airframe made of CFRP.

Flaps and other mechanical parts such as rings, ribs and stringers are made of aluminium.

Steel is used in the landing gear to withstand high stress of landing.

Titanium is used in the wing-fuselage connection as fasteners bolting the wings and fuselage together as well as the main pylon connecting the wing and fuselage (this is to make sure the wing has enough support due to sheer forces and bending moments). It's also used in the APU exhaust cone and some other parts across the aircraft.

14 Landing Gear

14.1 Configuration

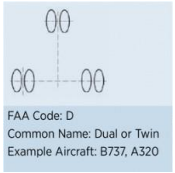


Figure 14.1

We have chosen the retractable tricycle arrangement for our design; it has been the most selected configuration for most Civil Aviation Aircraft because:

- The fuselage would be mostly level in static conditions which is favourable for ground operations and services such as passenger, cargo, and catering etc.
- Unlike early taildragger configurations, pilots will have an unobstructed view while on ground. The manoeuvring stability is also better as it has less tendency for nose to pitch down or ground loop (wing yawing to one side).
- Retraction of the gear can reduce drag during cruise conditions.

14.2 Clearance Angle

Aircraft fuselages are commonly designed to curve upwards near the tail section to avoid tail strike at high angles of attack while landing or take off. To avoid tail, strike the clearance angle needs to be larger than the maximum angle of attack.

$$\theta = \alpha_{max} + \frac{d\theta}{dt}.\Delta t$$

Where maximum angle of attack is estimated to be $\alpha_{max} \approx 15$, and nose rotation rate of $\frac{d\theta}{dt} \sim 3$.

However, it is important to note that during the climb out fuselage will move towards the runway. This effect is estimated as a time interval can be given as

$$\Delta t \approx \frac{300}{V_{off}}$$

Where $V_{off} \cong 144\ knots$

We can apply this to the equation, and we can find our clearance angle to be $21.25°$

14.3 Geometry

Based on the parameters we have provided in the PDR; we can begin to select the appropriate tire and wheel size. Let's first make the following assumption:

- MTOW to be 150000 lbs, 88% of load will be on main gear and 12% on the nose gear.
- Per FAA requirements, the load rating of the tyres must be 107% of the MTOW.

First, we distributed the load and found that each main gear tyre will require a rated load of 35310 lb and for nose gear 9630 lb.

Aircraft	Main Gear Tire Size/ply	Rated Load (lbs)	Nose Gear Tire Size/ply	Rated Load (lbs)
B737-700	H43.5 x 16-20/26	40,600	27 x 7.75R15/12	9650
B737-800	H44.5 x 16.5-21/28	44,700	27 x 7.75R15/12	9650
B737-900	H44.5 x 16.5-21/28	44,700	27 x 7.75R15/12	9650
B747-400	H49 x 19-22/32	56,600	H49 x 19-22/32	56,600
B767-200	H46 x 18-20/28	44,200	H37 x 14-15/24	26,700
B767-300	H46 x 18-20/28	44,200	H37 x 14-15/24	26,700
B777-200	50 x 20R22/26	45,200	42 x 17R18/26	36,100
B777-300	50 x 20R22/32	57,100	42 x 17R18/26	36,100
B787-8	50 x 20R22/34	63,050	40 x 16R16/26	
B787-9	50 x 20R22/34	63,050	40 x 16R16/26	
A320	46 x 17R20/32	46,000	30 x 8.8R15/16	14,340

Table 14.1

From the comparison table and estimates shown above, we decided the tyre sizes of 737-800/900 would be sufficient for our design. We'll verify them now by considering dynamic loads acting on the aircraft.

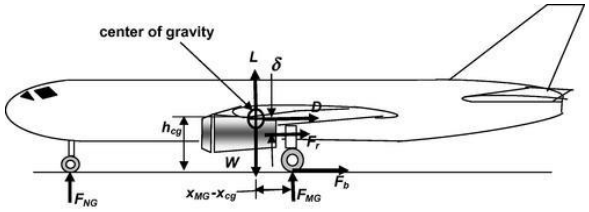


Figure 14.2: Free body diagram showing dynamic loads acting on an aircraft.

In FAA part 25.733 it states that *"The ground reaction of the tire corresponding to the most critical combination of airplane weight (up to maximum landing weight) and centre of gravity position combined with forces of 1.0g downward and 0.31g forward acting at the centre of gravity. The reactions in this case must be distributed to the nose and main wheels by the principles of statics with a drag reaction equal to 0.31 times the vertical load at each wheel with brakes capable of producing this ground reaction. This nose tire load may not exceed 1.5 times the load rating of the tire."* (Federal Aviation Administration, Department of Transportation, *Code of Federal Regulations (Annual Edition)* 2023)

We know by considering the force and moments during landing the maximum load on Nose Gear is given to be:

$$F_{NG_{max}} = W\left[\frac{(x_{MG} - x_{cg}) + \left(\frac{a}{g}\right).h_{cg}}{l_{wb}}\right]$$

Now let's consider the most stressing case and apply the following assumption.

- $\frac{x_{MG}-x_{cg}}{l_{wb}}$ to be $\frac{1}{8}$
- We can approximate MLW to be 0.839 of MTOW which gives 125850 lb.
- $\frac{a}{g}$ = 0.31 to satisfy FAR25.731 requirements for deceleration level.
- $\frac{h_{cg}}{l_{wb}} = \frac{1}{5}$

We shall transform the equation to be:

$$F_{NG_{max}} = 125850\left[\frac{1}{8} + 0.31 \times \frac{1}{5}\right]$$

We found the total force of 23533.95lb lb is applied on the nose gear, so the maximum load on each tyre would be 11767lb. Since 1.5 times of our tyre rating would be 14212.5lb so our tyre selection satisfies the requirement.

Since we have obtained the geometry of our tyres, we can now calculate our tyre and wheel weight using some extrapolations from trend analysis:

We estimate Tyre Weight to be:

$$W_{ti} = \frac{d_{ti} \cdot b_{ti} \cdot n_{plies}}{107}$$

And wheel weight to be:

$$W_w = 0.1\pi \cdot (d_w \cdot b_{ti} + \frac{d^2{}_w}{4})$$

Where $d_w \approx 1.4(Tire\ Rated\ Load)^{0.25}$

Since we have established some parameters for the landing gear, we can now begin to size our landing gear strut. We start by the required piston area given as:

$$A_h = \frac{F_s}{p_2}$$

Where F_s is the static load acting on landing gear and p_2 the internal pressure of piston assumed to be 1500 PSI.

When we have obtained our piston area A_h , we can find the inner diameter of our piston by using the area of circle equation. We can then size it up by 30% to obtain an estimate of the outer diameter of the piston.

The height of the landing gear strut is estimated to be $2 \cdot d_{ti}$.

We can now work out the volume hence the size of the landing gear strut assembly applying the dimensions above. Since our landing gear strut is made of Carbon Steel with density of 7840 kg/m^3), we can estimate our weight using the volume value.

	Main Gear	No.	Nose Gear	No.
Strut Assembly	849.2	×2	137.3	×1
Tyre	82.55	×4	14.1	×2
Wheel	76	×4	33.5	×2
Total Weight(kg)	2332.4		232.5	

Table 14.2

15. Economics

15.1 Market Gap

To assess the appeal of a new aircraft design, the consideration of design needs must be compared to with the market requirements. By identifying the gaps in the market, the marketing policies would be influenced in terms of expenses and costs which as a result, would determine its profitability and demand. The direct operating cost is a key indicator to achieving a viable aircraft financially. To achieve this, the Association of European Airlines (AEA) method was used to calculate the operating costs of the short-haul aircraft.

To evaluate the competitiveness of the concept aircraft, we identified the range various aircrafts that are being used in terms of turboprop and turbofan and compared them against the Pax. Figure 15.1 shows a trend analysis of these aircrafts and their ability to carry a certain number of payloads for a set range. From the data shown on figure 15.1, it is shown that the current commercial airliners mainly increase in range when the passenger number is increased. This forms a clear market gap, as there is no existing turbofan aircraft that ranges around the 1000km area with a passenger capacity of 190. This is highlighted on figure 15.1 and whilst this shows that there is no direct competitor, aircrafts like the Airbus A320 can be referred to as a competitor due to its 190 Pax capacity.

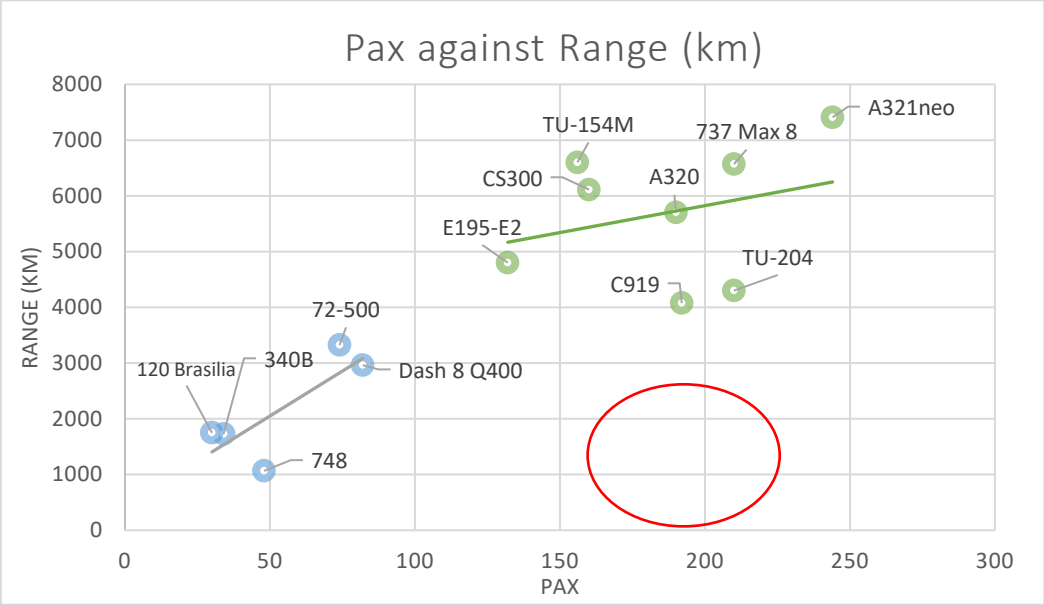


Figure 15.1: Pax against range graph showing market gap

15.2 Market Analysis of Intra-European Aircrafts

Air transportation is a key component of Europe's economic and social sector, playing a significant role in the connections between cities as well as job opportunities within the aviation industry. With a rough setback of the aviation economy during the pandemic, it is best if the market is analysed during its pre-pandemic level. The pandemic had caused the largest global recession since the Great Depression, which lead to a standstill in air transport. For this reason, it is necessary to analyse the market history before 2019, for a clearer understanding the industry as a whole. Figure 15.2 & 15.3 shows the overall social and economic benefits of air transport in Europe.

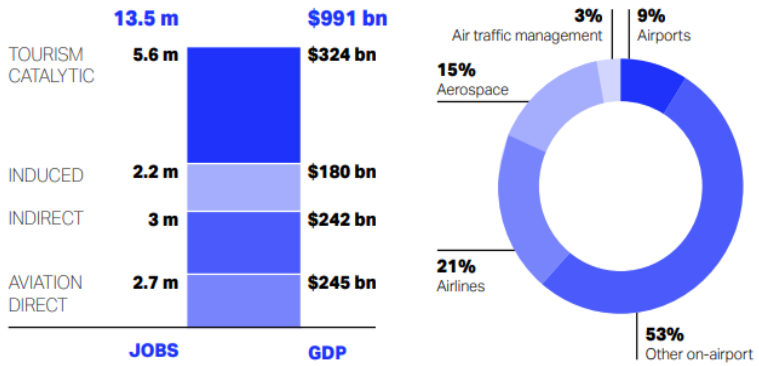


Figure 15.2: Economic and social benefits of air transport in Europe

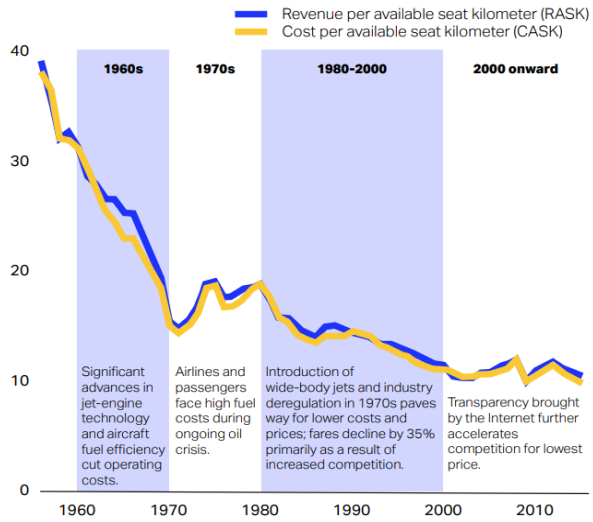


Figure 15.3: Cost of air travel over 60 years (figures deflated using US Consumer Price Index for transportation)

By 2018, the total number of jobs generated by air transport had reached 13.5 million, whilst boosting economic activity to $991 billion (figure 15.2). This is equivalent to 3.6% of the overall employment. As well as this, the cost of air travel has decreased by almost 75% since the 1960s due to the more fuel-efficient, cost-effective aircrafts that have been developed (figure 15.3). In conclusion, this shows the competitiveness of the current aviation market as airliners are competing for the lowest prices whilst reducing carbon emissions, resulting in a higher demand.

To analyse the aircraft market in terms of its compound annual growth rate (CAGR) up to 2042, we used the macroeconomics assumptions produced by Airbus. Overall, Airbus forecasts that the demand for passenger traffic will increase 3.6% annually between 2019-2042 (CAGR).

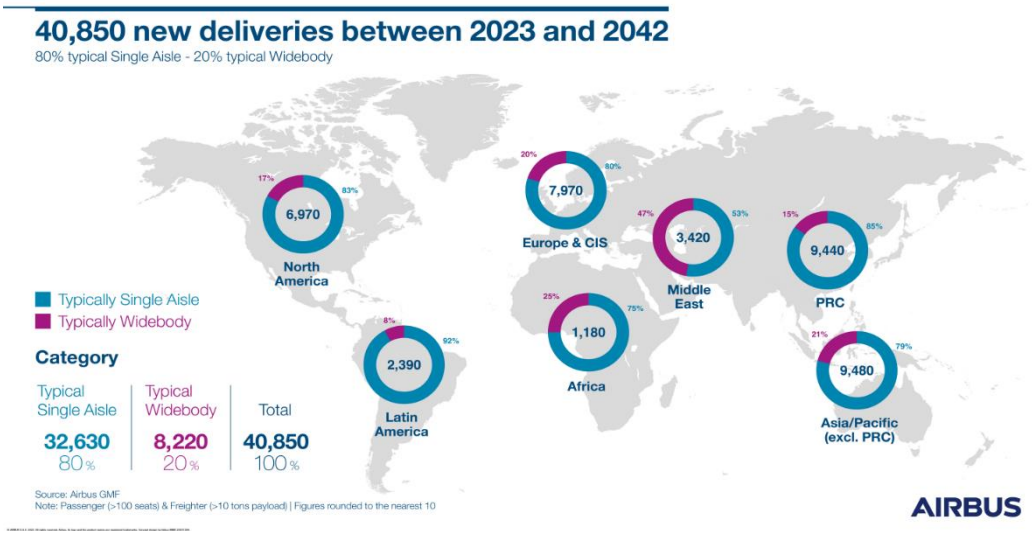


Figure 15.4: Economic and social benefits of air transport in Europe

Airbus assumptions are based on 100+ seater passenger aircraft, with both single aisle and wide-body formations. There is an 80:20 split between the two, outlying the demand for single-aisle.

The demand forecast shows that deliveries for passenger aircraft over the next 20 years will be around 7970, 6400 of which will be single-aisle aircrafts.

Passenger traffic forecast

Source: Sabre, Airbus
Note: CAGR based on non-oriented Origin Destination Number of Passengers

Traffic flow	2019-2032 CAGR	2032-2042 CAGR	2019-2042 CAGR
Domestic - Central Europe	1.5%	1.8%	1.6%
Domestic - Europe	1.3%	1.7%	1.5%
Domestic - CIS	2.9%	3.5%	3.2%

Table 15.1: Passenger traffic forecast between 2019-2042

Another statistic shown on the Airbus Global Market Analysis (table 15.1) is the passenger traffic forecast up to 2042. Domestic flights within different areas of Europe show a CAGR increase of up to 3.5%. This is based on 100+ seaters and 80% of which being single-aisle. The demand for air travel will no doubt, increase but to create an aircraft that will remain fuel-efficient whilst meeting customer needs, is the area of competitiveness that our concept aircraft faces. For this reason, this aircraft will proiritise the importance of minimal environmental impact, alongside a low direct operating cost.

15.3 Direct Operating Cost (DOC) Breakdown

To represent the direct operating costs for our aircraft, the Association of European Airlines (AEA) was used to analyse the financial, maintenance and flight costs. By doing so, a realistic account for the cost was developed in terms of the design of the aircraft and the technology. Table 15.2 provides the overall cost per seat-mile of the aircraft. Jenkinson's method was used to calculate the airframe cost, with Roksam's material cost estimates for a rough valuation.

	Concept Aircraft
Utilisation	1487
Manufacturers Study Price ($)	61,200,000
DOC seat mile cost (cent/seat-nm)	23.59
Total Cost per Trip ($)	11,797

Table 15.2: Overall DOC Calculations

15.4 Turnaround Time

A vital component towards the utility of the aircraft is the turnaround time. This is a part of the design phase that allows an early understanding of its profitability as the smaller the TAT, the higher the yearly utilisation leading to a lower cost per seat mile. The process includes the passenger and cargo handling, refuelling, cleaning, and catering. With our concept aircraft, there is a disadvantage that increases the TAT; the narrow body, single-aisle formation with 190 passengers means that the passenger loading time will be slightly longer than a wide-body aircraft. This would lead to lower utilisation but as the fuel efficiency is optimised instead, the long-term costs would be lower. Through comparisons of the TAT of competitor aircrafts (such as the Airbus A320 with a TAT of 44 minutes), the turnaround time for our concept aircraft was optimised to be 55 minutes. [Ref. 15]

15.5 DOC with Fuel Price Variation

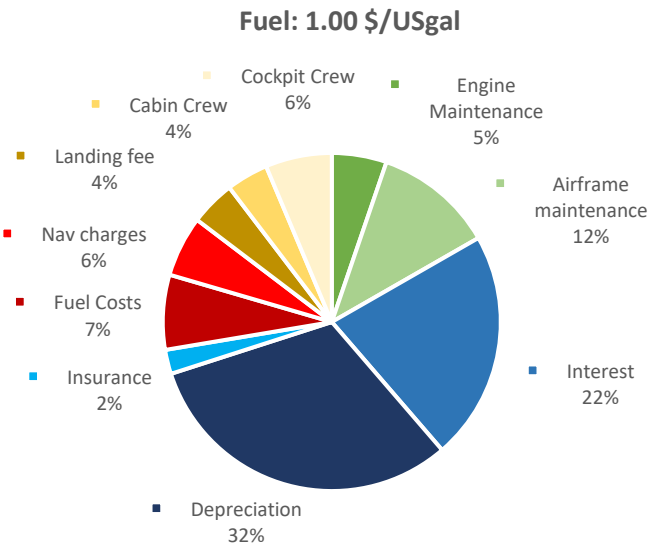


Figure 15.5: Pie Chart with 1.00 $/USgal

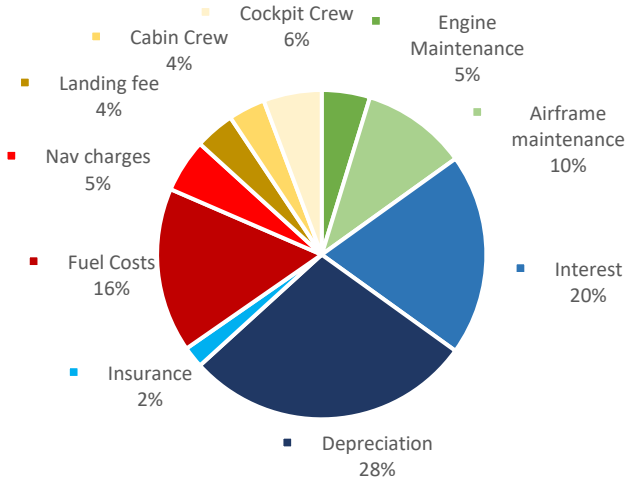


Figure 15.6 Pie Chart with 2.50 $/USgal fuel

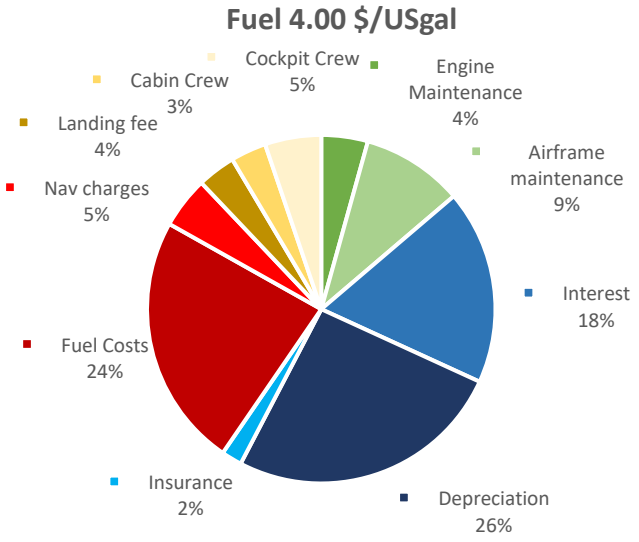


Figure 15.7: Pie Chart with 4.00 $/USgal fuel

Fuel Price	$/USgal	**1.0**	**2.5**	**4.0**
Financial Costs				
Depreciation	$/trip	3341	3341	3341
Interest	$/trip	2339	2339	2339
Insurance	$/trip	247	247	247
Maintenance Costs				
Airframe Maintenance	$/trip	1222	1222	1222
Engine Maintenance	$/trip	559	559	559
Flight Costs				
Cockpit Crew	$/trip	676	676	676
Cabin Crew	$/trip	427	427	427
Navigation Charges	$/trip	619	619	619
Landing Fees	$/trip	460	460	460
Fuel	$/trip	762	1907	3050
Total DOC Sector Cost	$/trip	10653	11797	12941
Total DOC Seat-Mile Cost	cent/seat-nm	**21.3**	**23.59**	**25.88**

Table 15.3: Detailed doc cost results

As shown in figures 15.7, 15.6, 15.5 and table 15.3, the DOC is impacted by financial costs; this includes the interest, depreciation, and insurance costs. The effect of fuel shown on the pie charts describes the volatility of the fuel market. Fuel costs can rise from 7% to up to 24% if the fuel price increases by 3 $/USgal. This is not a factor that can be modified and for this reason, must be observed.

15.6 Limiting Factors

An issue that arose during cost calculations is retrieving the competitor DOC costs as these values seem to have been kept in secrecy by the manufacturers, making it a struggle to compare the various aircraft prices. Without this, estimation costs become difficult to assume due to the lack of transparency. Another limitation that occurred was caused by out-dated sources such as Roskam's method. The report was created in 1985, which was an obstacle to overcome as the cost of materials were lacking in validity. This affected the airframe cost. With the dollar having an average inflation rate of 2.80% per year, there is an uncertainty in the costs that have been predicted. The calculations have considered inflation; however, the accuracy is limited.

15.7 Sensitivity Analysis

Calculating the DOC causes a few issues as the effect of future costs can drive the overall study price. The fuel price and the manufacturing study price (MSP) are two variables that are driven by the market force and not the manufacturer. For this reason, a sensitivity analysis was created with the two variables being reflected by the relative Pax-mile cost. The visual analysis shows that if the fuel price is $1 US gal with an airframe cost (AFC) of $50 million, the percentage variation decreases by 17% compared to if the fuel price was at $2.5 US gal at an AFC of $55 million.

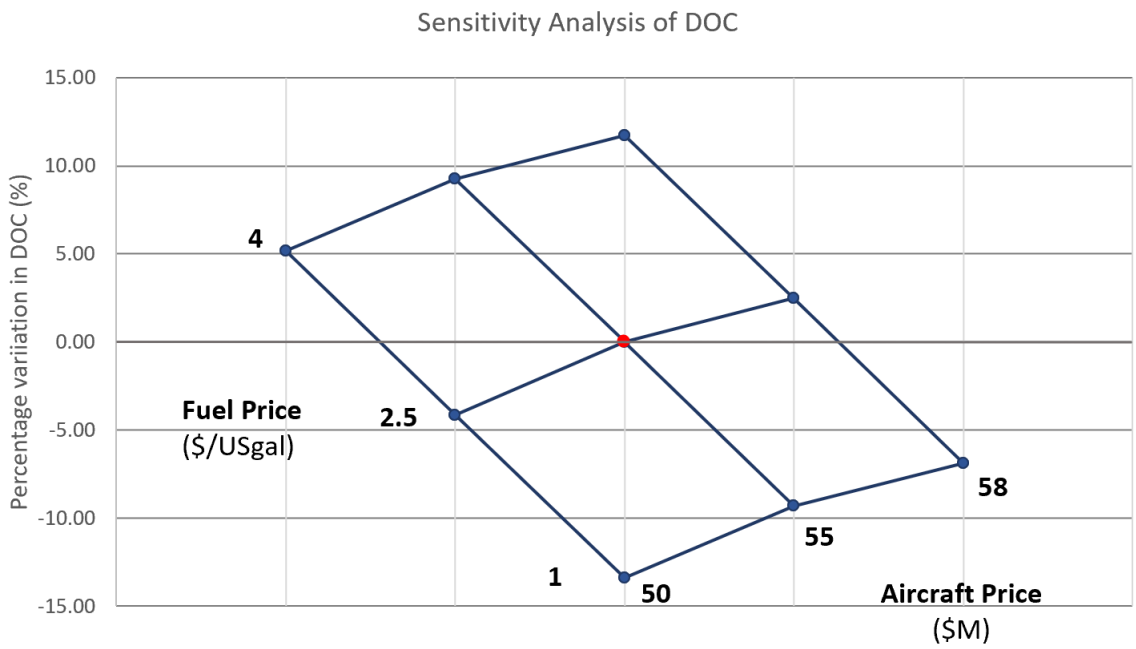
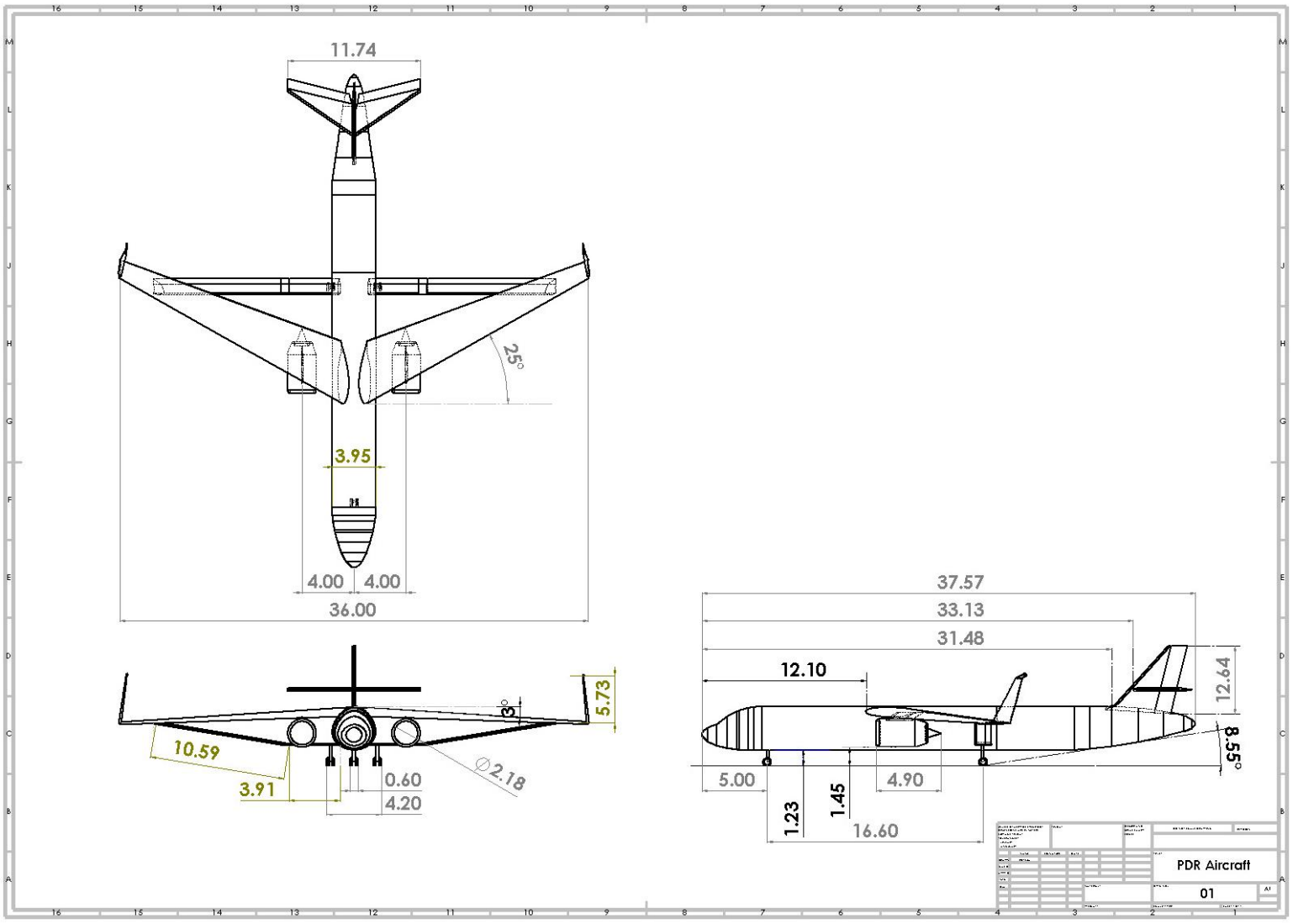


Figure 15.8: Sensitivity Analysis

16. General Arrangements Drawing and CAD Model





17. Avionics, Systems & Other Aircraft Considerations

17.1 IMA Avionics Architecture

This avionics architecture utilises a Side 1/Side 2 architecture. Figure 17.1 shows a simplified version of side 1 and side 2 would be a duplicate of the opposite side. The schematic shows us that on side 1 and side 2, there is a crosstalk and interchange of data. Sometimes, a third ADIRS (air data and inertial reference system) may be installed as a backup for the first two.

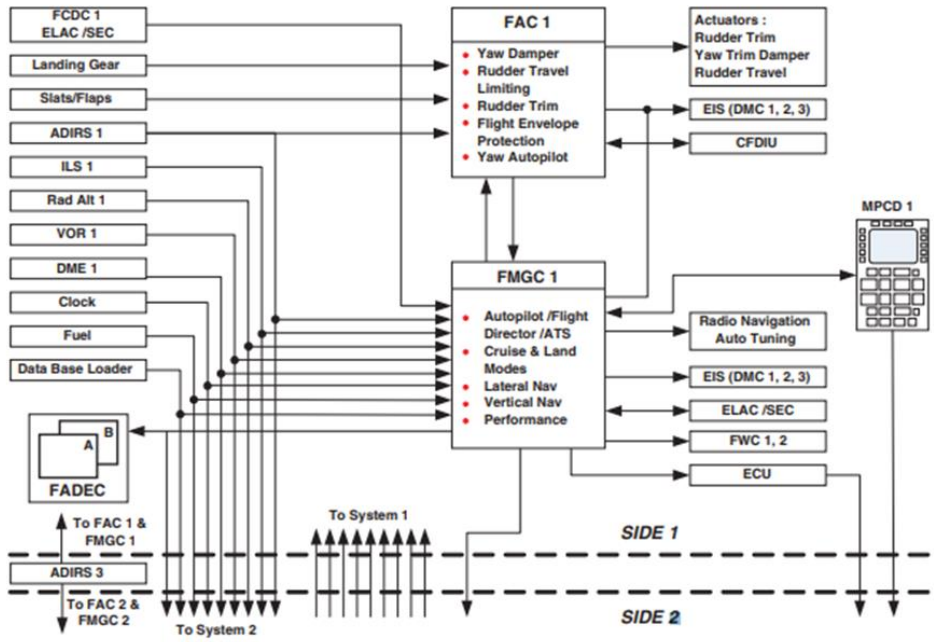


Figure 17.1. Airbus Avionics architecture (Moir, Ian) [Ref. 12]

There are two components in the centre of the architecture. One of these components is the Flight Augmentation Computer (FAC). This provides functions linked to the yaw channels such as yaw damper, rudder travel limiting, rudder trim, flight envelope protection and yaw autopilot as shown in Figure 16.1. The airframe systems such as the landing gear and the slats/flaps will send inputs to the FAC. From this, it will recognise any changes to the aircraft configuration notably during the take-off and landing. The FAC will move the most appropriate actuators to use the yaw control functions.

The other component is the Flight Management & Guidance Computer (FMGC). The component receives inputs from the fly-by-wire system and the radio navigation aids (ILS, Rad Alt, VOR and DME) then analyses this data, generates, and tunes accurate navigation information including the aircraft's current position, heading, altitude, speed, and flight profile to the pre-loaded flight plan.

17.2 Fuel System

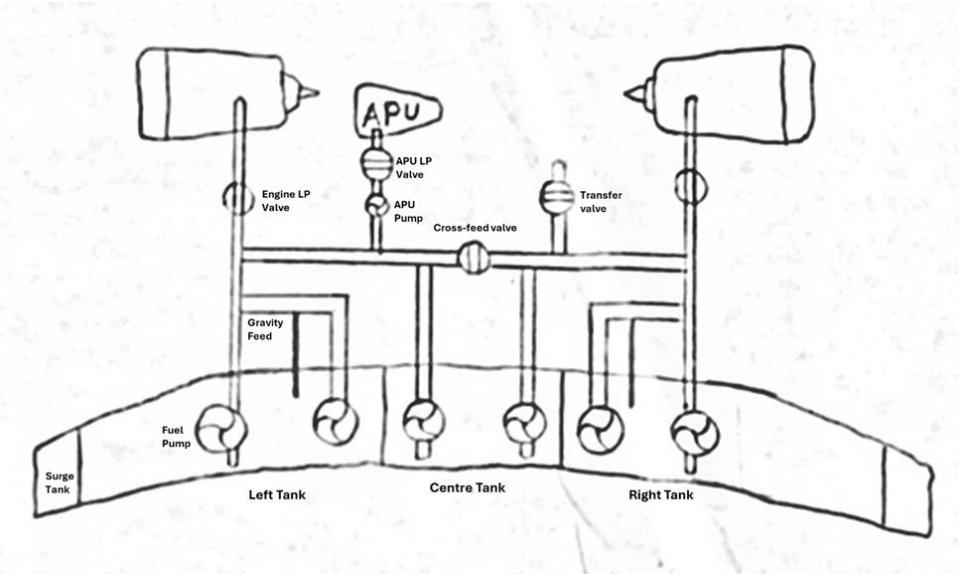


Figure 17.2. Airbus Fuel System Schematic (Airbus Technical Notes. (n.d.)) [ref. 13]

Figure 17.2 shows a schematic of the fuel system that will be used on our aircraft. It is comprised of a fuselage-integrated centre tank, and tanks on each wing, along with vent surge tanks. Vent surge tanks serve to counteract thermal fuel expansion without cockpit indications.

Overpressure protectors are integrated into wing and vent tanks, with the centre tank featuring a protector linked to the left-wing tank.

In total, there are six fuel pumps, two in the centre tank and two on each side of the wing tanks. At standard conditions, these pumps will supply fuel to the engine and APU if necessary. A double motor cross-feed valve unifies the fuel system, with the option to override through the X FEED button.

In scenarios of pump failure, the wing tanks use suction valves for gravity-fed fuelling. The centre tank pumps have automatic and manual modes, influencing fuel sequence and pressurization. The pilot can determine potential failures using the ECAM FUEL page monitor pump status which displays amber boxed LO for low pressure. An amber box with a cross line indicates a manually switched-off pump.

Refuelling is automated but can be manually controlled. A de-fuel/transfer valve facilitates fuel system maintenance. The Fuel Quantity Indication (FQI) system, comprising probes, densitometer, and a Capacitance Index Compensator, provides ECAM data and controls refuelling. Manual measurements are possible using magnetic fuel level indicators. Fuel temperature exceeding limits triggers a master caution. A shroud drain mast under the fuselage manages leaked fuel. This sophisticated system ensures optimal fuel management, distribution, and safety throughout the aircraft's operations.

This fuel system works for our aircraft compared to others as this design looks to use a more automated approach to the controls similar to our competitor Airbus aircraft. Whereas other aircraft such as the Boeing 737 emphasise on maintaining a high level of pilot control.

17.3 Electrical Systems

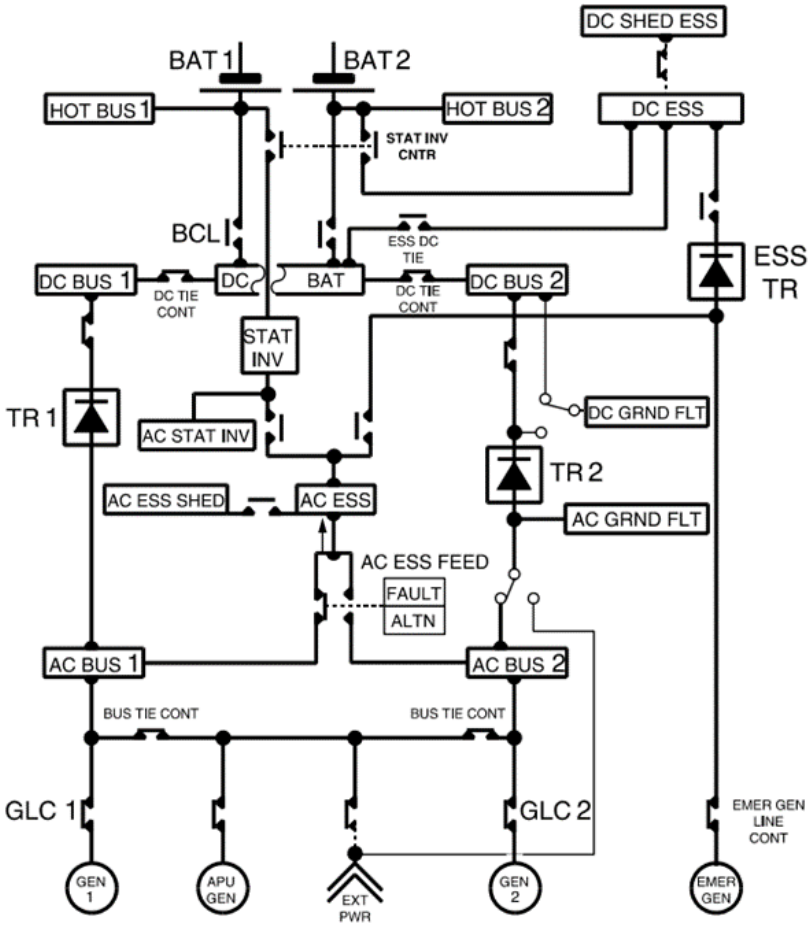


Figure 17.3. Electrical system Architecture

The aircraft's electrical power generation system is a sophisticated configuration designed to ensure a reliable and redundant power supply in various operating conditions. The primary sources of AC power comprise two engine-driven AC generators, each boasting a nominal power of 90kVA, alongside an auxiliary power unit (APU) AC generator with an equivalent power rating. In the event of a main generator failure, an emergency generator, hydraulically driven by the Ram Air Turbine (RAT), automatically deploys, providing a backup power source with a nominal capacity of 5kVA. On the ground, a dedicated ground connector delivers a power output of 90kVA. The DC network is supported by three identical Transformer/Rectifier Units (TRU), with two normally in use and the third serving as a backup in emergency configurations or in case of (Transfer Rectifiers which change 115 volts AC in 28 volts DC) TR 1 or TR 2 failure. Additionally, the aircraft is equipped with two batteries, each boasting a 23Ah nominal capacity. While on the ground, these batteries primarily facilitate autonomous APU starting. In emergency scenarios, they play a crucial role in powering specific equipment during Ram Air Turbine deployment and after landing gear extension, providing an additional layer of resilience to the aircraft's power distribution system.

This system is very similar to our competitor Airbus Aircraft. This electrical system is a more suitable fit for our aircraft compared to other systems such as one used in the Boeing 737. One of the main differences between this system and a Boeing 737 is that it uses a centralized bus architecture compared to a split bus architecture. We are looking to keep the aircraft as light as possible which this system architecture can provide. Brady, C. (n.d.). [Ref. 14]

18 Conclusion

This report demonstrates an in-depth analysis of a preliminary design for a short-range commercial aircraft. The aircraft achieved and went beyond our desired requirements. Our validated design is optimised for low fuel burn and low operating costs, coinciding with the industry's growing commitment to minimising air travel's environmental footprint. We have not only created high-performing aircraft but have also established the groundwork for a more environmentally responsible future in aviation by methodically addressing numerous design drivers and critical criteria.

The aircraft boasts a windowless and narrow fuselage with a six-abreast seating plan, emphasizing structural efficiency and passenger comfort. Its wings feature struts, a high aspect ratio, and a 25-degree sweep angle for optimal performance with a high L/D ratio on short-haul flights. Powered by a strategically placed turbofan engine, the design prioritizes efficiency for speeds of Mach 0.77-0.79, enabling shorter journeys and increased daily flights. With a six-abreast seating plan and a T-tail configuration, the aircraft minimizes weight, enhances fuel efficiency, and aligns with the anticipated growth in the narrow-body aircraft market from 2023 to 2030. This design approach aims to deliver a balance of efficiency and passenger satisfaction while addressing future market trends.

This report also shows how the aircraft was improved beyond the requirements through multiple iterative processes. The aircraft went through 5 iterations including aerodynamic performance which adjusted the cruise Mach number to prioritise fuel efficiency, contrail reduction, and enhanced passenger comfort. In addition, it also iterated the structural analysis which varied the strut placement to refine the structural integrity.

To summarise, our aircraft incorporates aerodynamics, propulsion, stability, and control, as well as modern system design and smart material choices, encompassing all elements for economic and technological competitiveness. Furthermore, by using eco-friendly materials and aerodynamics, the design exceeds existing sustainability criteria, putting it at the forefront of environmental responsibility in aircraft technology.

19 Beyond Final Design

19.1 Alternative Propulsion

In a world that’s looking for new ways to become more environmentally friendly, as well as more efficient, the aeronautical world looks to introduce hybrid aircraft into the industry. Boeing, alongside NASA, have being looking into this concept over recent years, and have developed a concept of hybrid aircraft up to Phase 4 of the development stage.

Hydroelectric engines have selected as a modification that can be made to the aircraft due to pollution, both environmental and noise as well as potential for an increase in aircraft performance. It’s believed that the use of hybrid engines will reduce the noise pollution down to 32 – 42 dBA compared to aircraft today operating around 60-85 dBA . This both benefits the passengers by providing a more comfortable and enjoyable flight experience, as well as residents who will be happier with the lower during ground operation and take-off.

NOx emissions are one of the key factors in the formation and persistence of contrails. During combustion the engine releases nitrogen oxides which is a significant contributor to acid rain and the depletion of the ozone layer. By using hybrid engines have the potential to eliminate NOx emissions, as shown in by the table below. It can be observed that the use of a more powerful electric engines, nearly eliminate the emission produced.

Mid-Cruise Emissions (CFM56-7B Baseline)			
	1380 HP ICAC Constrained	1750HP ICAC Inactive	7150HP ICAC Inactive
Absolute NOx Reduction	73.5%	73.9%	99.3%

Table 19.1, NOx emissions due mid cruise across different electric motors17

With the use of fuel cells, and batteries the weight of the engine is likely increase. This will negatively impact MTOW, studies of a similar aircraft show that the designed aircraft's MTOW will increase potentially by nearly 10,000 kg (ref), meaning the engines will need to be scaled up, and other variables will change. It will likely cause issues with wing stability. As the aircraft in this design as a relatively thin wing with a long span, the implementation of heavier engines can increase wear on wings, we well as increase bending moments. To counter this the material of which the wings are made from may have to be altered or the addition of stiffeners to help increase the bulking resistance.

Not only does this effect the performance of the aircraft but may increase manufacturing costs and maintenance of the wings. The costs of the aircraft may be due to more variables such as fuel cells and additional motors, needing to be monitored and repaired.

The use of a hybrid aircraft could lead to a reduction of fuel burn by 56% for a 900-mile flight[16], by utilities our current design of a strut-braced wing, whilst using conventional fuel. This therefore means that the use of aircraft isn't limited to the reduction of fuel, and the number can be further increased by burn alternative fuels.

19.2 Windowless Fuselage

Demand in the aviation industry as called for lighter and more aerodynamic planes. One modification that can be that reaps weight savings, manufacturing simplicity and aerodynamic advantages is the implementation of a windowless fuselage.

However, with removing the windows opens to a huge trade-off between customer satisfaction and operational needs. A potential to compensate for this could be the use of virtual reality headsets or the use of LCD panels. Both these solutions offer a simple, yet low weight alternative however the question remains is how does this effect the cost of production of the aircraft?

As the fuselage of this aircraft is based around the Airbus A320, the study by the CEAS applies to this concept idea. The weight of a windowless fuselage can be considered by the following[18]:

$$W_{skin\ only} = D * L * t * \rho_{alloy} * \pi$$

$$W_{near\ hole\ reinforcement} = 2 * L * \left(t_{reinforcement} - t\right) * \rho_{alloy}$$

$$W_{window\ metal\ frame} = \sqrt{2} * A_0 * k * b_{semiminor\ axis\ of\ window} * \rho_{alloy} * N_{window}$$

$$W_{alloy\ removed} = 2 * a_{semimajor\ axis} * b_{semiminor\ axis\ of\ window} * N_{window} * \rho_{alloy} * t$$

$$W_{montor\ camera\ and\ systems} = (W_{monitor} * N_{window} + W_{camera} * N_{cameras}) * X$$

$$W_{trad_fuselage} = W_{skin\ only} - W_{alloy\ removed\ to\ make\ windows} + W_{window\ metal\ frame} + W_{near\ hole\ reinforcement} + W_{windows} = 2155.4kg$$

$$W_{windowless\ fuselage} = W_{skin\ only} + W_{montor\ camera\ and\ systems} = 1560kg$$

Further data suggests, the final weight savings of a windowless fuselage is 750 Kg. This is calculated by using the available data resulting in around 600 Kg weight savings. However, during the design phase each kilogram can factor 1.25 Kg in the total plane weight give use our final weight savings. This allows us to reduce MTOW having a positive knock-on effect on the amount of fuel need to for the mission resulting in less fuel consumption and lower costs for the airline.

19.3 Other Considerations

Another modification that could be to the aircraft is the production of a slimer, but a slightly longer fuselage. Opt-in for a narrower fuselage result in a smaller airframe leading to a lighter aircraft, with more aerodynamic benefits.

On the aerodynamic side, there aircraft will have a lower profile, reducing the skin friction drag of the fuselage therefore resulting in a lower total drag. Reduced drag means that less thrust from the engines during take-off and cruise, therefore decreasing the fuel consumption of the aircraft and the operational costs.

Weight reduction not only means better fuel efficiency and improved performance but also means less material need to manufacture the fuselage. This Leaders to lower manufacturing costs due to less materials required but also a shorter production time as there is less of the aircraft too build. As a result, the aircraft has a competitive advantage other competitors due to a short lead time, meaning the aircraft can be in operation quicker.

To ensure maximum efficiency from the addition from the strut there are key aerodynamic features that can contribute to the performance to optimise the wing. The use of struts increases the vortexes caused around the wing-strut, and strut-fuselage junction, as there is more points of contact. A method to reduce vortex and reduce flutter on the wings, novel aerofoil shapes can be implemented around the junctions to alleviate the fast acceleration of flow to delay/eliminate shockwaves.

Other modifications that can added to reduce these the affects can be through novel bump-like features over the lower surface of the wing. This in turn reduces the pressure peaks present of the lower surface[33].

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Appendix

Aircraft data sheet

Parameter	Value	Unit
Wing Area	112.59	m^2
Wingspan	47.45	m
Aspect Ratio	20	
Wing thickness (t/c)	0.12	
Wing Taper Ratio	0.4	
Sweep @ Quarter Cord	17.2	degrees
MAC	2.52	m
Strut MAC	1.55	m
CLmax @ Landing	2.074	
Distance of LE @ root from nose	12.1	m
Tail Area	27.5	m^2
Tail Span	11.74	m
Tail AR	5	
Tail TR	0.256	
Tail Chord Root	0.6785	m
Tail Chord Tip	0.1737	m
Tail MAC	1.78	m
Fin Area	31.9	m^2
Fin Span	12.64	m
Fin AR	1.82	
Fin TR	0.303	
Fin Chord Root	0.6072	m
Fin Chord Tip	0.184	m
Fin MAC	3.2	m
Number of Engines	2	
Position of Engines from fuselage	4	m
Thrust per Engine	211600	N
Engine Pod Length	4.9	m
Engine Pod Diameter	2.18	m
Landing Gear Layout	Tricycle (Retractable)	
Main Landing Gear Tyre Size	H44.5/18.5/24	D(inches)/W(inches)/Ply
Nose Landing Gear	H26.5/10.5/12	D(inches)/W(inches)/Ply
Angle of Rotation	21.25	Degrees
Main Gear Geometry	0.94*2.26	Width*Height
Nose Gear Geometry	0.53*1.34	Width*Height
Cruise Mach Number	0.78	Mach
Initial Cruise Alt	31000	feet
L/D	15.96	
Take-Off Length	2127	m
Approach Speed	0.22	Mach
Landing Length	1150	m
Fuselage Length	37.57	m
Height of Fuselage Datum above Ground	1.23	m
Fuselage Diameter	3.77	m

Cabin height	2.1	m
Position of CG from Nose	17.6	m
Manufacture Study Price	61.2	$million
Engine Price	5.6	$million
Aircraft DOC (2.5$/Usgal)	23.59	cent/seat-nm
Turnaround Time	55	minutes
Refuel Time	15	minutes
Utilisation	1487	trips per year